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Performance, Safety, and Engineering Trade-offs of Rust and C++ in Nokia RAN Control-Plane Systems

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Tiivistelmä — Referat — Abstract

Radio access network (RAN) control planes are written mostly in C++ for performance, at a well-known cost in memory safety and potential for concurrency bugs. Rust promises comparable performance with compile-time safety, but its fit for this latency-critical setting is unproven.

The thesis compares modern C++ and Rust using a user equipment (UE) Context Management service that keeps per-device state, as a 5G base-station control plane (gNB-CU-CP) does. Average-throughput comparisons leave four dimensions open: tail latency, the defect classes Rust prevents at compile time, safety-model overhead, and incremental adoption.

One language-agnostic specification is fixed and then implemented in both C++ and Rust. The two versions use the same algorithms, thread layout, memory allocator, and build settings. Where C++ and Rust cannot match line for line, the difference is written down and kept in the measurement. Both versions pass the same parity tests. The benchmark then raises the request rate step by step until the service overloads. The evaluation also adds fault injection and a foreign function interface (FFI) prototype.

Safe Rust refuses to compile code that contains any of the seven injected memory-safety and data-race defects, whereas C++ compiles the same code. Both languages compile code that contains seven further defects, such as deadlocks.

Each version was run with two memory allocators, the system default and `jemalloc`. Before overload, Rust answers a typical request up to 1.5 μs faster than C++ on the end-to-end round trip. Only 0.250% of that trip is spent in Rust or C++ code. Most of the rest is thread handoff. On a same-host clock that defers coordinated-omission fill past the timed window, Rust still finishes more calls at 784 thousand requests per second with both allocators. That clock times schedule drift, not service time, and it understated Rust's throughput. On further hosts, outside the tested family, most of the lead comes from the queue. Matching the handoff leaves Rust 0.6% to 1.3% ahead on the system allocator. With several requests in flight on a further host, the wait is shorter for Rust with `jemalloc` and for C++ with the system allocator.

On all three machines tested, Rust uses a few MiB more peak memory than C++ with both allocators. Existing C++ code can call the unchanged Rust service through a thin C interface. The crossing adds at most 10.91% of one signalling call's measured cost.

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List of acronyms

Each acronym below is also written out where the text first uses it. Terms that are not acronyms, such as *tail latency* and *shard*, are collected in the list of key terms that follows.

Acronym	Expansion
3GPP	3rd Generation Partnership Project, the body that specifies the 5G standards
ABI	Application binary interface
AI	Artificial intelligence
AI-RAN	An architecture in which AI and radio access network workloads share one accelerated infrastructure
API	Application programming interface
ASN.1	Abstract Syntax Notation One
CPU	Central processing unit
CVE	Common Vulnerabilities and Exposures
CWE	Common Weakness Enumeration
DPDK	Data Plane Development Kit
EU	European Union
FFI	Foreign function interface
FIFO	First in, first out
FNV-1a	Fowler–Noll–Vo hash function, variant 1a
GCC	GNU Compiler Collection
gNB	Next-generation NodeB, the 5G base station
gNB-CU	gNB central unit
gNB-CU-CP	gNB central unit control plane, the domain this work sits in
gNB-CU-UP	gNB central unit user plane
gNB-DU	gNB distributed unit
HDR	High dynamic range, of the latency histogram
IMT-2030	International Mobile Telecommunications-2030, the ITU framework for 6G
ITU	International Telecommunication Union
KPI	Key performance indicator
kops/s	Kilo-operations per second
KVM	Kernel-based Virtual Machine
MPSC	Multi-producer, single-consumer
NG-RAN	Next-generation radio access network
PHY	Physical layer
RAII	Resource Acquisition Is Initialization
RAN	Radio access network
RQ	Research question
RRC	Radio Resource Control
RSS	Resident set size
UE	User equipment, an end device such as a phone or a modem
UECM	UE Context Management, the subsystem reimplemented and measured here

Key terms

The table below gives plain meanings for the network, the service, the timing, and the safety words used in the thesis.

Term	Meaning
Network	
Control plane	The messages that set up and control a call. The user's data travels on the user plane
User plane	The path that carries the user's data. The control plane sets that path up and takes it down
Signalling	The messages that create, change, and end a connection
Handover	Moving one connected device from one cell to another, so the call can continue
Cell	One coverage area of the radio network
Mobility	A connected device changing cell while the call continues
UE context	The stored record for one device. A handover or a release changes that record
Bearer	A data channel for one device's user data. The UE context lists its bearers
Reconfiguration	A change to a connection that stays up, such as adding a data channel
Release	Ending one device's connection and deleting its record
Procedure	One action a person would notice, such as setting up a call or moving between cells. It is made of many smaller steps
Open-RAN	The same split base station, opened so each part may come from a different company and run as ordinary software
5G core	The network behind the base station. It forwards each call onward
The service	
Shard	One slice of the device records. Exactly one thread owns that slice
Producer	A thread that sends the test's requests and waits for each reply
Worker	The thread that owns one shard and carries out the requests for it
Bounded queue	The queue in front of a shard. It has a fixed size and waits when full, so a service that cannot keep up shows delay rather than dropped requests
Operation	One request for one device, such as setup or release
Variant	One of the four builds under test. The four are C++ or Rust, each linked with one of the two memory libraries
Allocator	The library that hands out memory and takes it back
Timing	
Tail latency	How long the slow calls take, rather than the average call. This thesis reads that from the p99, the p99.9, and the p99.99
Median	The middle call, once the calls are ordered by how long they took. Half are faster and half are slower
Histogram	The record of how long each call took. Tail latency is read from its slow end
Open-loop	The sender keeps to its send schedule and does not wait for each reply

Term	Meaning
Offered load	The rate the test tries to send, counted whether or not the service answers in time
Achieved rate	How fast the service finishes work. It matches offered load until saturation
Throughput	How many requests the service finishes each second
Saturation	The load at which the service can no longer keep up and requests start to wait
Overload	Load past saturation. Requests wait because the service cannot keep up
Round-trip	The time from the planned send until the reply comes back. It includes the wait in the queue
Bare metal	The program runs on the computer itself, with no virtual machine between it and the hardware
Hypervisor	Software that runs virtual machines. The reported campaign has none between the test and the hardware
Guest	A virtual machine an earlier measurement ran in. The reported results come from the bare-metal computer
Userland	Programs and libraries above the kernel. The container supplies the compilers, and the host supplies the kernel
Safety	
Data race	Two threads use the same memory at once and at least one writes, with nothing to keep the two uses in order
Race condition	A bug that depends on which of two steps finishes first. Safe Rust still compiles these
Use-after-free	Using memory after it has been given back
Double-free	Giving the same memory back twice
Use-after-move	Reading from a value after its contents have been moved elsewhere. The old name is still valid but empty
Ownership	Rust's rule that each value has one owner. A read of a record whose owner is already gone does not compile
Borrowing	A temporary loan of a value. The compiler rejects a loan that lasts longer than the value
Safe Rust	Code outside `unsafe` blocks, where ownership, borrowing, and aliasing are checked at compile time
Unsafe	An `unsafe` block or function where the programmer takes responsibility and the usual checks do not apply. C++ has no such mark
Sanitiser	A tool that watches a running C++ program and reports some defects. It can report only a path a test actually runs
Undefined behaviour	An operation the C++ standard gives no meaning, such as signed integer overflow. The compiler may assume it never happens
Static analyser	A tool that inspects source code without running it. This study ran none

1 Introduction

Mobile networks are moving from specialised hardware to software on ordinary servers. Functions that once ran on fixed-purpose, vertically integrated equipment are now reimplemented as programs on general-purpose machines. The radio access network (RAN), long the most hardware-centric part of the stack, is part of that shift.

In 5G, the base station, formally the next-generation NodeB (gNB), is split into a central unit and one or more distributed units. The central unit is split again into control-plane and user-plane functions [1, 2]. The control plane carries the signalling, the messages that set up and control the call. The user plane carries the phone's data. Open-RAN pushes the same idea further: the RAN becomes virtualised, multi-vendor software linked by open interfaces [3]. The control plane that results, the gNB central unit control plane (gNB-CU-CP), is a large, stateful, highly concurrent program. It tracks every connected device. The standards call each device a user equipment (UE). The control plane runs Radio Resource Control (RRC) signalling [4] and coordinates mobility and handovers, all within strict latency limits.

A single site can run thousands of connection-management and mobility procedures per second. Each procedure has to complete within the millisecond-scale budget the 5G system sets for control-plane signalling [5]. A deployed control plane may track millions of active devices at once. As the RAN moves to cloud-native, disaggregated deployments, the correctness, reliability, and timing of this software increasingly set the quality of the whole network. An outage here is not a local defect. Neither is a tail-latency spike: a few calls take much longer than the rest, even when the average still looks fine. Every device the site serves is affected [6, 7].

This control-plane software is, today, written mostly in C++. The choice is a practical one: C++ offers the low-level control, mature tooling, and performance that latency-critical systems demand. But that performance comes with a well-known cost. C++ places the burden of memory and concurrency management on the programmer, and decades of experience show that this burden is routinely mishandled. The typical failures are use-after-free errors, double-frees, data races, and related defects. They are expensive to find, dangerous in production, and responsible for a large share of security vulnerabilities in industry practice [8, 9]. Rust has emerged as the first mainstream systems language to promise an escape from this trade-off: memory safety and data-race freedom enforced at compile time on machine-checked foundations [10, 11], with performance intended to

be competitive with C++ [12]. Whether that promise holds for a telecom-grade, latency-critical control plane is not well understood. Predictability under overload matters as much as raw speed there. This thesis, carried out at Nokia, addresses the question directly. It develops two functionally equivalent implementations of a representative RAN subsystem, a UE Context Management service that keeps per-device state, as a gNB-CU-CP does, in modern C++ and in Rust. It compares them under a realistic synthetic signalling workload.

Three terms recur throughout, and the first is the *gNB-CU-CP*, the control-plane class this work sits in. *UE Context Management (UECM)* is the compact service reimplemented and measured. *Tail latency* is the third. More than average speed, it decides whether such a system meets its latency budgets [6, 7].

1.1 Motivation

Three trends make this comparison timely.

- **Cloud-native RAN.** Control-plane functions now run on general-purpose servers, so the software is longer-lived, more concurrent, and must meet its latency budgets without the determinism of purpose-built hardware. Open-RAN disaggregation turns the base station into virtualised, multi-vendor software [3]. A latency regression or a memory-corruption defect is then a network-wide risk, and the language choice is an engineering decision.
- **AI on the RAN host.** Artificial intelligence (AI) used to be only a tool. It helps to choose radio settings, such as how much power to send. That tool does not have to run on the radio's own server. Now AI programs also run on that server, next to the radio software [13]. Both run at the same time, so they share one processor. International Mobile Telecommunications-2030 (IMT-2030) is the 6G framework of the International Telecommunication Union (ITU). It lists AI together with communication as one named job for 6G [14]. The architecture forming around that job is AI-RAN, in which AI and radio work share one accelerated server [15]. A control plane there no longer has the host to itself. It shares the central processing unit (CPU), memory, and cache with inference work it does not schedule. When that neighbour takes capacity, the control plane saturates sooner: it stops keeping up with demand at a lower load than it would alone. Chapter 5 therefore drives both implementations past that point. It also separates the time spent passing work between threads from the time spent doing the work itself.

- **Memory-safe languages.** Large software organisations report that the majority of their serious security vulnerabilities stem from memory-safety errors, and that adopting memory-safe languages reduces them [8, 9, 16, 17]. Independent work argues that memory safety now needs standardised requirements if it is to be procured at all [18]. Cybersecurity agencies urge a move away from C and C++, and the European Union (EU) Cyber Resilience Act adds legal duties from September 2026 and December 2027 [19, 20, 21]. The Act names no programming language. Direct evidence for a RAN control plane is still missing. Providing it is the motivation for this work.

1.2 Industrial context

This thesis is carried out at Nokia. The company is one of the small number of vendors that build, maintain, and evolve production RAN software end to end. The five largest suppliers took 96% of worldwide RAN revenue over the first three quarters of 2025, with concentration by Herfindahl–Hirschman Index at a ten-year high [22, 23]. The subsystem this thesis studies, UECM, is a compact model of the per-UE state work a gNB-CU-CP does.

In the architecture of the 3rd Generation Partnership Project (3GPP), the gNB-CU-CP is the control-plane half of the central unit. It hosts RRC and manages the UE context lifecycle [1, 4], under the millisecond-scale budgets the 5G system sets for control signalling [5]. Chapter 2 places that control plane in the split 5G base station, the gNB. Because it is written in C++, it inherits that language's performance and its memory-safety and concurrency hazards. The three trends of Section 1.1 meet here. Software of this class runs on live operator networks. The language choice is a production concern.

UECM is small enough to reimplement faithfully and reason about in full. It still carries the properties that make control-plane code hard. The implementation follows a public, fixed specification, not proprietary code, so readers outside Nokia can reproduce the comparison.

A decision with a generational horizon. A language choice in software of this class outlives the generation it was made for. 3GPP has allocated Release 20 to the 6G studies and Release 21 to the normative work. The ITU asks for technology proposals in early 2029 and for specification submissions by mid-2030 [24]. That work covers migration from 5G and interworking with it. A control plane written now is maintained alongside the next generation, not retired. IMT-2030 still names security and resilience as a design principle

and still carries the ultra-reliable low-latency scenario in strengthened form [14].

The measurements here use a current-generation subsystem and published requirements. No claim is made about requirements that have not been written.

A concrete failure mode. A moving train full of passengers stands for the connected UEs at one site. A handover moves one passenger from one cell to another, and the call can continue. Each passenger has a stored record, the UE context. When many passengers move at once, the control plane is under a mobility storm. One passenger leaves, so that record is deleted, while a message for that passenger is still being handled. Which of those two steps finishes first is a race condition. If the read comes after the delete, it is a use-after-free. In C++ this still compiles. Tests that do not hit that ordering can pass. When many calls run at once, the bug can corrupt another passenger's record. That race can crash the thread that is serving the call. A few calls can become much slower than the rest. That jump in the slowest calls is a tail-latency spike. In safe Rust the same design does not compile: ownership and borrowing reject the read of a record that has already been deleted. That is why this thesis measures both *safety* (which defects are prevented, and when) and *predictability* (tail latency under overload), not average throughput alone.

1.3 Problem statement

Two properties of this domain make the language choice consequential. First, manual memory and concurrency management in C++ exposes the software to defects such as use-after-free, double-free, data races, and iterator invalidation. These defects are silent at compile time and costly to find and fix at telecom scale. Second, what governs whether RAN latency budgets are met is not average throughput but worst-case and tail behaviour under load and overload [6], a property that is easy to regress and hard to measure correctly. Rust promises to remove the first class of defect at compile time while claiming competitive performance on the second, but that claim rests mostly on general-purpose benchmarks rather than on stateful, concurrent, latency-critical telecom workloads.

Bringing these threads together, the concrete problem this thesis addresses is the following. Telecom vendors are under growing pressure, both technical and increasingly regulatory, to adopt memory-safe languages. For the RAN control plane they still lack *direct, controlled* evidence on which to base that decision. It is not established whether a Rust implementation of a stateful, concurrent control-plane subsystem can match modern C++ in *tail latency and predictability under overload* (as opposed to average throughput). It is also

unknown whether Rust's compile-time guarantees actually eliminate the specific defect classes that affect the existing C++ code, and what incremental adoption *costs* inside a large C++ codebase. The available evidence is largely general-purpose and average-case [12, 25]. For a representative subsystem, four questions therefore stay open:

- **Predictability:** tail latency and behaviour under overload, not average throughput alone.
- **Prevented-defect scope:** which C++ defect classes Rust removes at compile time, and how the rest are caught.
- **Resource overhead:** throughput, CPU use, and memory under the safety model.
- **Migration cost:** effort, foreign-function boundaries, and risk inside a large C++ codebase.

Answering them under controlled, reproducible conditions is the problem this thesis sets out to solve.

1.4 Research questions and hypotheses

The thesis is organised around four research questions. They put latency predictability first, then defect scope, then resource cost, then incremental adoption:

RQ1 How do Rust and modern C++ compare in latency predictability (median, p99, p99.9 and p99.99 tail latency, and the maximum) for a RAN control-plane workload under increasing load and overload?

RQ2 Which classes of memory-safety and concurrency defects that occur in C++ are eliminated at compile time in the Rust implementation, and at what stage and cost are the remaining ones detected?

RQ3 What is the overhead, if any, of Rust's safety model (bounds checks, ownership-driven copies, abstraction cost) on throughput, CPU utilisation, and memory behaviour?

RQ4 What are the practical implications (effort, foreign function interface (FFI) interoperability, and risk) of adopting Rust incrementally within an existing C++ RAN software ecosystem?

Three hypotheses are written down before any measurement.

H1 For this workload the two languages are performance-equivalent below saturation, with any difference only in the far tail.

H2 Safe Rust prevents at compile time the memory-safety and concurrency defect classes that C++ only tries to catch after the program is built.

H3 Calling the Rust service from C++ through the FFI adds only a small share of one signalling call's time. The real cost of adding Rust one piece at a time is engineering work, not slower calls.

Chapter 6 revisits all three hypotheses against the facts.

What RQ2 covers. The compile-time safety claims of RQ2 apply only to *safe Rust*, the code outside `unsafe` blocks, where the compiler can enforce its rules (Section 2.2). The measured service core is written in safe Rust, and so is every injected defect but one, which bypasses the checks inside `unsafe` on purpose (Section 5.3.4). Chapter 4 explains how the adoption prototype keeps `unsafe` code limited to its C application binary interface (ABI), the layer where C++ calls into Rust.

The RQ2 list has fourteen defect classes. Safe Rust is expected to stop seven of them at compile time. It is expected not to stop the other seven. That split is the boundary the study looks for, not only a check that the guarantee works. The seven boundary classes are these:

- lock-order deadlock
- lost update
- reference-cycle leak
- integer overflow in a release build
- worker-failure blast radius
- double-free inside `unsafe`
- forgotten lock guard

A previous study already estimated that boundary from a list, not from programs that were built. It took the Common Weakness Enumeration (CWE), a long list of bug types. It marked each type as one that safe Rust would stop or would not. It then counted real product bugs in 68 open-source C projects. 58.2% of those classified bugs sat in types that safe Rust would have made almost impossible [26]. That share is a best case: it assumes the new code stays in safe Rust. A type counts as stopped if the mapping says so, even if no compiler has rejected a program. RQ2 asks the same question of programs that are built and run. That is how it can say when the types still open are caught, and what it costs to catch them.

1.5 Contributions and outline

The thesis leaves three artefacts for reuse: a fixed, language-neutral specification of the service, a C++ and a Rust implementation that pass the same parity tests, and a timing harness that reaches past saturation. The first of its results compares latency and resource use below and past saturation, and separates the core operation from the handoff around it. The second maps where Rust's compile-time guarantee stops, using defects built on both sides of it. The third bounds what one call from C++ into Rust costs. The discussion turns these into guidelines for adding Rust one piece at a time.

Chapter 2 places the work in the 5G base station and among earlier comparisons, and introduces the two safety models and how tail latency is measured. Chapter 3 gives the method for each research question. Chapter 4 describes the two implementations, and Chapter 5 the machines, the workload and the results. Chapter 6 answers the research questions, revisits the hypotheses and states the limitations, and Chapter 7 concludes.

2 Background: Systems Languages for Latency-Critical Software

2.1 RAN control-plane systems and the gNB-CU-CP

This thesis compares modern C++ and Rust in the next-generation radio access network (NG-RAN). In this network, user equipment (UE) moves between cells, generating signalling events such as handovers. These events can occur frequently and place a high load on the network. At the same time, signalling procedures must complete within strict time limits, as defined by the key performance indicators (KPIs) in the 5G service and RAN specifications [5, 27].

NG-RAN software is continuously developed and must handle many operations at the same time. This makes it important to understand not only how quickly operations usually complete, but also how slowly they can complete under heavy load. For example, a handover must finish within a specified time limit. If most handovers finish quickly but a few take too long, the average latency may still look good even though the time limit is exceeded for some handovers. These delays can affect the users involved and cause the KPI to be violated.

For this reason, this thesis focuses on tail latency, which describes the delays experienced by the slowest operations. By measuring tail latency, the comparison can reveal performance problems that average latency alone may hide.

The gNB is split on two axes. The first split is central versus distributed. The gNB runs a protocol stack: a set of layered radio protocols from the air interface up to RRC signalling. A central unit (gNB-CU) hosts the higher layers. One or more distributed units (gNB-DU) host the lower layers. Those layers sit close to the radio and have strict low-latency requirements. The second split is control versus user plane, inside the central unit. It splits the control-plane function (gNB-CU-CP) from the user-plane function (gNB-CU-UP). These parts talk over standard links. Uu is the radio link from the device. NG goes toward the 5G core, F1 sits between the CU and the DU, and E1 sits between the CU-CP and the CU-UP [1, 2, 28]. Open-RAN pushes the same links further. Each part can then come from a different vendor and run as software on ordinary machines [3]. Figure 2.1 shows the part studied in this thesis, UE Context Management (UECM), inside the gNB-

CU-CP. UECM owns the per-UE control-plane state. It also handles RRC signalling and mobility. These are integral control-plane functions, which makes UECM a representative stateful subsystem. It does not stand for the control plane as a whole.

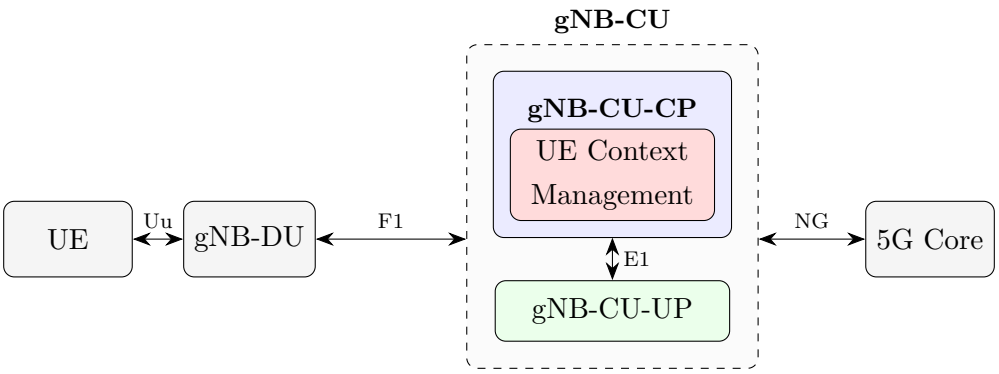


Figure 2.1: Where the studied subsystem sits in the 5G NG-RAN. UE Context Management is highlighted inside the gNB-CU-CP.

2.2 Memory safety and latency measurement

Rust separates code into *safe Rust* and code that uses `unsafe`. In safe Rust, the compiler checks how memory is used, including ownership and borrowing, to prevent many common programming errors [10, 11]. When code uses an `unsafe` block or an `unsafe` function, the programmer must make sure that certain operations are used correctly, because the compiler cannot check all their safety requirements. Unsafe code is often needed when Rust calls C code through the C ABI or uses raw pointers. C++ has no keyword that marks such code.

C++ manages many resources through Resource Acquisition Is Initialization (RAII), in which an object releases what it holds when it goes out of scope. Nothing in the language stops a program from keeping a pointer to that object afterwards, or from sharing it between threads without a lock. C++ projects therefore catch them with tools that watch a running program. This study runs three of them, AddressSanitizer, ThreadSanitizer and UndefinedBehaviorSanitizer (Section 3.4). A sanitiser can report a defect only on a path a test actually runs, so a clean report covers the tests and not the program.

How a test sends its requests decides whether its slow calls are counted. A test that waits for each reply before sending the next request slows down whenever the service does. It then sends fewer requests during a slow spell and records fewer slow calls than users would see. A test that keeps to a fixed send schedule avoids most of this [29]. It can still fall behind its schedule, so HdrHistogram adds a delay sample for each request that should have been sent during that time [30]. Section 5.2 describes how the harness adds How each research question isthese samples, and Section 5.3.2 shows that when they are written can change a result.

2.3 Research gap

The RAN control plane has many threads and tight latency bounds. Tail latency under too much load, not mean throughput, sets the quality the network delivers. C++ gives the control and speed the domain needs. But it defers detection of whole classes of memory and concurrency bugs to runtime. Rust promises to block those classes of errors at compile time at no run-time overhead. It also offers a step-by-step adoption path through the C application binary interface (ABI).

Five vendor-hosted Master's theses, all done at Ericsson, have compared candidate languages for RAN and telecom software. One compares run-time performance with development effort for C++, Rust, and Go [31]. One tests whether Rust can replace C in RAN node management [32]. One builds a packet processor in async Rust, with one thread per core, and measures it against the Data Plane Development Kit (DPDK) program it mirrors [33]. One measures median and p99 latency for control-plane steps across five async frameworks [34]. The last builds matched C++ and Rust versions of one program. That program turns a packed Abstract Syntax Notation One (ASN.1) message back into a readable form. It compares them with statistical tests [35]. All five are directly relevant. They do not all favour Rust. Olsson and Stenström's packet processor reaches a fifth to a half of DPDK's throughput, with far worse latency [33]. Arvidsson and Kaller's Rust p99 rises more steeply past a threshold rate [34]. Karlbäck and Orö pick C++ where speed is critical. They still find Rust lightest on peak memory [31]. Olsson's management software is as fast as C or faster. It uses more peak memory [32]. Emberg and Wadsten's Rust version is comparable to C++ in one session. It is 85–91% worse in the next [35].

Three of them change the language together with the run-time and the thread model, so a measured gap cannot be blamed on the language rather than on the framework [31, 33, 34]. In that packet-processor study the DPDK program does two to five times as much work [33]. A compiler does not move throughput that far. The run-time does. Olsson's management study and Emberg and Wadsten do not change the run-time with the language [32, 35]. But they report run time and peak memory, not tail latency under rising load. Emberg and Wadsten come closest in method to this thesis. Their program keeps no state between calls. The service in this thesis stores state and uses many threads. Figure 2.2 places the five vendor theses beside the general-purpose comparisons [36] and the matched systems-software pairs [37, 38]. Each of these groups leaves out at least one thing the figure asks for. No group keeps the design fixed, measures the tail, and drives the service into overload.

The place these studies come from is part of the gap too. All five studies come from one vendor. Their conclusions have never been tested against another firm's software, targets or house rules. So what is known about this question is what one firm's frameworks say about it. The published record on the other side is thinner. There is no study of Rust among the indexed papers of the vendor hosting this work. The nearest peer-reviewed stance from that side argues for making memory-error sanitisation practical. It does not argue for replacing it with a compile-time guarantee [39]. That is the strategy this thesis tests against.

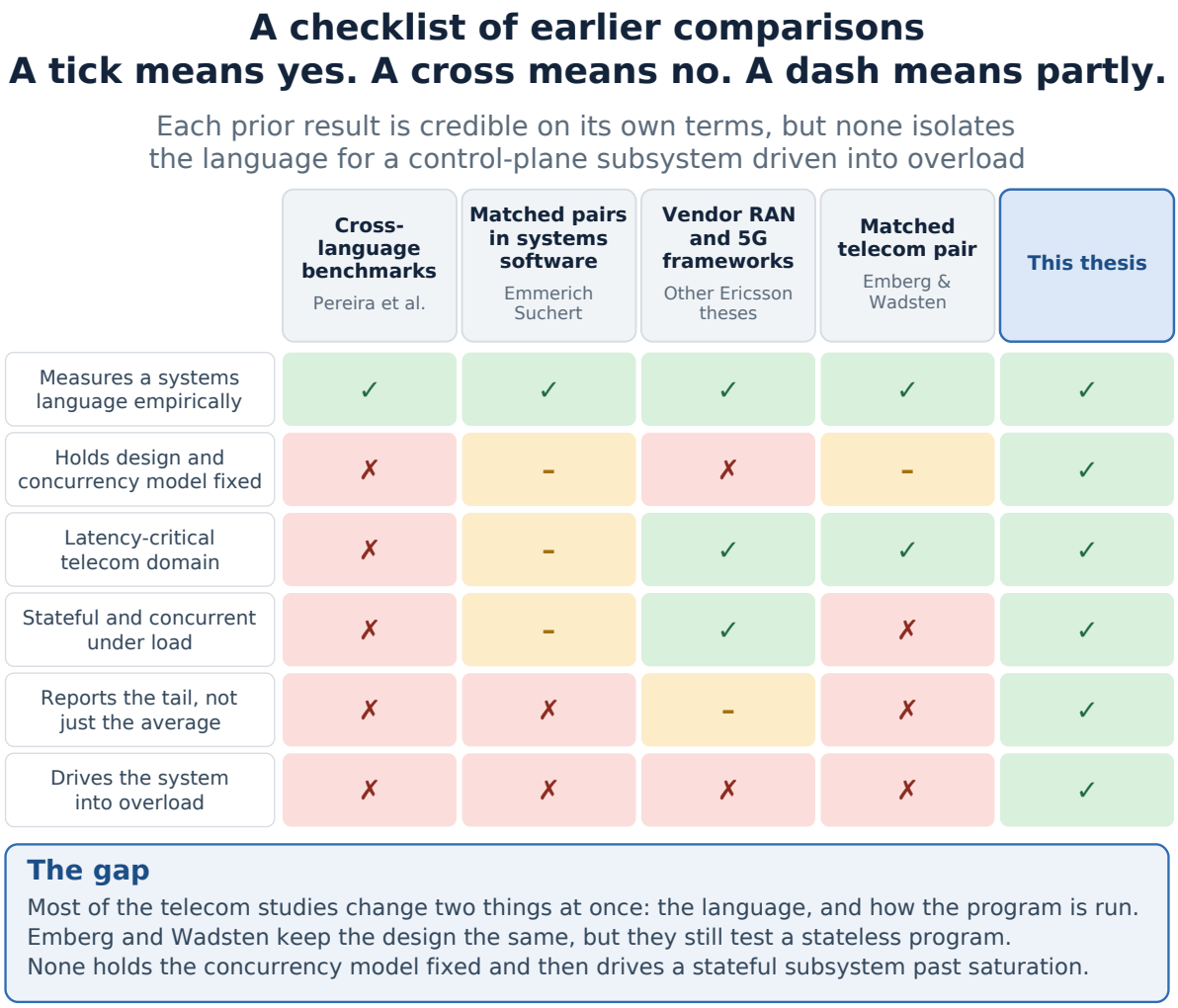


Figure 2.2: A checklist of earlier comparisons. Each row is a property a controlled answer would need, and each column is a group of studies. The vendor-RAN column is the other Ericsson theses. Emberg and Wadsten have their own column because they keep the design the same in both languages.

What is missing from prior work is a direct, controlled test of a typical control-plane part. The open questions are these. Does Rust hold its tail latency under too much load against modern C++ when the thread model, operations, and data layout stay fixed? Which bug classes does it remove, and at what stage are the rest caught? What does its safety model cost in throughput and in CPU and memory? What does step-by-step adoption cost in practice? Closing this gap is the purpose of the four research questions. The next chapter grounds them in the part under study, the bug classes, and the timing at stake. It also sets out the method.

3 Research Methodology

The previous chapter said what earlier studies leave out. This chapter says how the study closes that gap.

3.1 The UE Context Management subsystem

The part under study is a UE Context Management service. Its operations follow the UE-context lifecycle, RRC procedure execution [4], and mobility and handover that the 3GPP design assigns to the control-plane part of a split gNB (Section 2.1). Figure 3.1 shows the UE-context lifecycle both versions encode the same way. The mixed workload of Section 5.2 walks each UE through that machine.

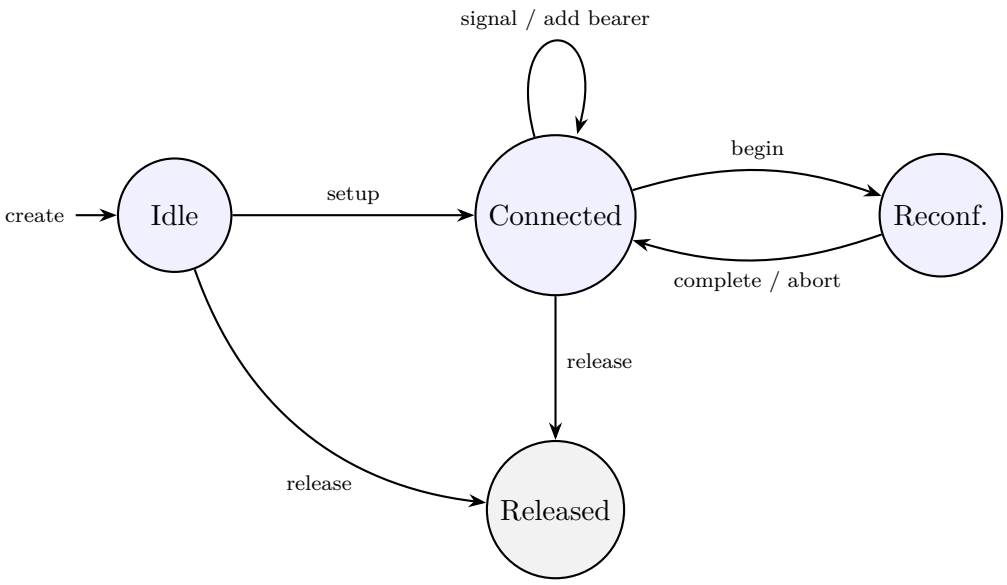


Figure 3.1: UE-context lifecycle state machine. Operations invalid for the current state return an error instead of changing state.

UECM is core RAN layer-3 work. It has the traits that make live control-plane code hard. Each UE has state that changes, the operation lifecycle is strict, and many UEs run at once. A delay target is the longest a control-plane procedure may take. The 5G service requirements name latency as one such limit [27]. For the move from a battery-saving idle state to the start of sending data, that limit is 10 ms [5]. The procedure is a chain of steps. It is late if the chain is late, even when most steps are fast. Findings about how a language handles those traits are most relevant to other stateful, multi-thread control-

plane services of this kind. The part is also small enough to specify in full and build twice. That is what makes a parity-tested comparison possible.

The interface is small and built around the lifecycle. It can create and release a context, set up a connection, handle an in-session signalling message, add a data radio bearer, begin, complete, or abort a reconfiguration, and look up a context's state under load. Each operation is defined for each state of Figure 3.1. An illegal operation returns a typed error and leaves the context unchanged. That closed set lets the same tests and the same workload drive both versions along the same paths.

As Section 1.4 states, RQ2 asks which defects safe Rust stops at compile time, and which ones it does not. Section 3.2 sets out the catalogue. For each defect, the expected result was written down before the programs were run. Section 5.3.4 reports what each one did.

The study reads tail latency as well as the typical call. Two facts make that the right choice.

The first fact is what a phone user waits for. Setting up a call, or moving a phone to the next cell, is not one step. It is a chain of control-plane steps, and they have to finish in order. The user is still waiting until the slowest step in that chain is done. A busy site runs thousands of these chains every second. A call that is slow only rarely therefore still happens often enough to notice.

The second fact is about the test clock. The sender is open-loop: a late reply does not hold back the next request. While offered load stays below saturation, achieved rate matches it and delay stays small.

When the timetable is made faster, a limit appears and the delay jumps. That limit can be the service, or work the sender still does while the timer is running, such as writing down the late replies. Because either one can cause the jump, the study sends more work than the service can finish and then checks which of the two the timer was counting.

3.2 The defect catalogue

The defects considered in this study fall into two broad families: *memory-safety defects* and *concurrency defects*. This classification is useful because the two families capture different ways in which C++ control-plane code can fail, and because safe Rust is expected to provide different guarantees for them.

Table 3.1 lists the 14 classes injected in Chapter 5, each in the form it would take in UE Context Management, with the outcome expected of safe Rust before the study was run.

Section 5.3.4 and Figure 5.9 report what was measured.

Table 3.1: Injected defects and their predicted outcomes. Buffer overflow is listed but not injected.

Defect	Example in UECM	Result	Safe Rust
Memory safety			
Use-after-free	Handler uses a context after it was released	Crash or corruption	Prevented
Double-free	Release frees the same context twice	Crash or corruption	Prevented
Use-after-move	Request uses a context after it was moved	Empty identity	Prevented
Buffer overflow	No applicable scenario in the context store	n/a	Not injected
Concurrency			
Iterator invalidation	Bearer list changes while it is being iterated	Crash or corruption	Prevented
Data race	Two workers update the same field at once	Wrong value	Prevented
Unguarded shared mutation	Multiple workers mutate the same context	Wrong value	Prevented
Thread-unsafe handle	Non-atomic handle is shared between workers	Corrupted refcount	Prevented
Defects safe Rust does not automatically prevent			
Lock-order deadlock	Two handovers acquire locks in opposite orders	Hang	Not prevented
Lost update	Two workers overwrite each other's updates	Wrong value	Not prevented
Reference-cycle leak	Contexts keep references to each other	Memory leak	Not prevented
Integer overflow	A bearer's tunnel identifier overflows a signed 32-bit integer	Wrong value	Not prevented
Worker failure	Worker fails while holding shard state	Wider outage	Not prevented
Double-free across FFI	C++ releases a Rust handle twice	Crash/corruption	Not prevented
Forgotten lock guard	Lock is not released after an early return	Hang	Not prevented

3.3 Research approach

The study runs experiments on a matched pair of versions. It builds that pair, plus the harness that drives them. The harness is the program that sends the load and records the times. Both are built from one fixed spec. Evidence comes from three separate methods: a load sweep contributes to RQ1 and RQ3, controlled fault injection answers RQ2, and

an FFI migration prototype answers RQ4. A weakness in one method does not weaken the others. How much the host can move the tail, for example, does not change the fault-injection result.

Each question asks how one outcome depends on one variable, the language. The language has to be the only planned gap between the two objects under test. That rules out the most common way to compare languages. In that form two programs written on their own are measured, and the gap is taken to be the language. Such a gap also carries the design, the libraries, and the skill of their authors. Building both sides from one spec removes that confound. Where a question can be answered with numbers, it uses numbers. Where it cannot, the answer is still by a fixed plan, in words. That is how this study treats the kind of guarantee in RQ2. RQ4 asks about the design cost of the application programming interface (API).

The closest prior work is the five vendor-hosted theses of Section 2.3. The wider literature came from the digital libraries of the Association for Computing Machinery and the Institute of Electrical and Electronics Engineers, the proceedings of the USENIX Association, and Google Scholar. The searches combined the terms *Rust*, *C++*, *memory safety*, *tail latency*, *systems programming*, *performance comparison*, and *RAN* or *5G control plane*, and citation chaining extended them. The five theses appear in none of the citation databases, neither Crossref, OpenAlex nor Semantic Scholar, so chaining cannot reach them. They were found by searching the university repositories directly, the Digitala Vetenskapliga Arkivet portal for the Nordic universities and Finna for the Finnish ones. Finna returns no thesis on Rust in a radio-access or telecom setting. An affiliation search of the vendor's indexed works in OpenAlex finds one that mentions Rust, an unrelated preprint. No search returned a study that measures the tail of a Rust and a C++ version of the same stateful, multi-thread control-plane part under too much load.

Table 3.2 is the map for the rest of this chapter. Each row is one research question. The columns say which test answers it, what was built for that test, which data the test produces, and which number is reported.

RQ1 asks how long a call takes, and RQ3 asks how much processor time and memory the service uses. H1 is the hypothesis for both questions. It says the two languages are performance-equivalent below saturation, with any difference only in the far tail. One hypothesis then has to hold for the delay and for the resources together, which is why that wording is its main weakness. The two need not move together. Chapter 6 returns to that wording.

RQ2 records whether the compiler accepts the program and what the C++ sanitisers

report. These results are independent of the machine. Most of RQ4 consists of sizes or counts, which are likewise machine-independent. Build time is measured with a stopwatch. A faster machine reduces both the Rust and C++ build times, so absolute times depend on the machine used. To control for this, the study reports the ratio of the two build times, measured sequentially on the same machine. Thus, the comparison is made within the same machine, rather than across machines.

Table 3.2: How each research question is answered. RQ1 and RQ3 use two methods. The first measures the full round trip, from sending a request to its completion. This shows the delay that a real deployment would experience below saturation. The second separates the core work from the handoff. This shows how much of the delay comes from the language itself. Past saturation, extra delay samples are written after the timed window.

RQ	Method	Artefact / data	Primary metric
RQ1	Load sweep to overload, interleaved repeats	Benchmark harness, latency histograms	median, p99/p99.9/p99.99, and maximum vs. offered load
RQ3	Same sweep, resource capture	`getrusage` samples per run	peak resident set size (RSS), CPU, throughput
RQ1, RQ3	Core called directly, then through the service	Matched microbenchmark pair	operation vs. cross-thread handoff
RQ2	Controlled fault injection	Defect catalogue, three C++ sanitisers, `rustc`, Loom/Shuttle, Miri	compile-time / runtime / undetected
RQ4	FFI migration prototype	C++ host + Rust C-ABI boundary	boundary size, `unsafe` scope, per-call cost

3.4 Methods per research question, in outline

RQ1 and RQ3 use an open-loop load sweep from a light load to well past saturation [29]. Four builds take turns inside each repeat, so a noisy moment hits all four.

The time from send to reply mixes two costs. The core operation is the work the language does on the device record. The handoff passes that work to the worker thread and brings the answer back. A second timing measures those two costs apart. The round-trip spends most of its time on the handoff, so the split shows how much of the measured time the language's own work can account for.

A third timing measures the bounded queue and the reply handle on their own, without

any `ShardCore` work in the path. The two languages build these pieces differently, so this test focuses only on the submit-and-wait synchronisation between the producers and the worker. Several producers compete for the same bounded queue, and each waits on a one-shot handle until the worker signals back. Together with the core-versus-handoff split above, this helps narrow down which part of the path may contribute to the overload delay.

Past saturation, extra delay samples are written after the timed window. They record the wait without adding recorder work while the timer is running.

RQ2 is answered by controlled fault injection [40]. Each bug class is a small program of its own, written in both languages. The C++ side is built and run under AddressSanitizer, ThreadSanitizer, and UndefinedBehaviorSanitizer. Those three tools watch a running C++ program for memory bugs, thread bugs, and undefined behaviour. Undefined behaviour is a mistake the language does not define, so a running program may do anything. The safe-Rust side is compiled. Where it compiles, it is run with a time limit, and the run is stopped if it takes too long. No static analyser is used, on purpose. A static analyser reads the source without running it. The C++ columns report the tools this study ran. They do not claim that no other C++ tool could find more.

RQ4 is answered by a working prototype in the direction a real migration takes. A C++ host calls an unchanged Rust core across a C ABI. That thin layer converts between the C++ host and the Rust core. The per-call cost is measured on a real operation and not inferred from an empty crossing. Memory bugs that cross a language sit in that layer [41].

3.5 Fairness as a controlled variable

A language comparison is only as good as what it keeps the same. The spec is fixed before either version exists. Several choices are then matched on both sides, and never tuned to favour one language. Those choices are the thread model, the allocator and its arena count, the opt level, the hash function, the histogram method, and the load schedule. The opt level is how hard the compiler tries to make the code fast. An arena is a pre-sized pool the allocator draws memory from. Both sides use the same speed setting and the same pool count. Where the two languages could not be written the same way, the gap is written down and left in place.

The two sides are shown to match by comparing their logs, operation by operation. For a given seed they also run the operations in the same order. A seed is the starting number of that random order, so the same seed repeats the same sequence. The four builds take

turns on the same pinned cores inside each repeat. A pinned core is reserved for the test, and no other work is allowed to use it. Running the same work one build after another reduces differences caused by changes over time in the processor's recent memory and branch guesses. The cache is the small fast memory next to the processor. A branch guess is the processor's guess about which way a condition will go. The test neither empties the cache nor lets the processor practise those guesses before each build. The work and the cores stay fixed, but the processor's recent memory and branch history do not.

3.6 Statistical treatment

The statistical analysis plan was decided before the experiment was run. The plan first specified which comparisons would be tested. These comparisons could not be changed after seeing the results. In particular, a load could not be added to the statistical analysis because its results looked interesting in the load sweep.

For each planned comparison, the study uses an exact paired permutation test over 20 interleaved repeats. The test is *paired* because C++ and Rust are compared within the same repeat. This means that both languages are measured under the same conditions at approximately the same time. For example, if the system is temporarily slower during one repeat, that slowdown affects both measurements.

The permutation test asks whether the observed difference between C++ and Rust could have occurred by chance. To answer this, the test repeatedly swaps the C++ and Rust labels within the paired measurements. It then checks how often a difference as large as the observed difference occurs after these swaps. A small p-value means that a difference of this size would be unusual if there were no real difference between the languages.

For each comparison, a bootstrap confidence interval is also calculated [42]. The confidence interval shows the uncertainty around the measured difference. It indicates the range of differences that are reasonably compatible with the observed repeats.

Because several comparisons are tested, the Holm correction [43] is applied. Without a correction, testing many comparisons increases the chance that at least one comparison appears statistically significant simply by chance. The Holm correction accounts for the fact that several tests are performed.

There is a limit to how small the p-value can be with a fixed number of repeats. Before the Holm correction, this limit is 0.0000019. After the correction, it is 0.000027. These limits are reported so that a reader can distinguish between a result showing no detectable difference and a result where the test does not have enough resolution to distinguish a

smaller p-value. In this study, a *tie* means that the test found no statistically detectable difference between C++ and Rust.

Not every reported measurement is statistically tested. Some measurements are included only to describe the behaviour of the system.

Throughput at the selected load is not tested because the experiment itself fixes the request rate. The test sends requests at the selected rate, so this value is already determined by the experimental setup.

Jitter measures how much latency varies from one call to the next. It is reported as a descriptive measure because it is not part of the research questions.

Maximum latency is also not tested. The maximum is determined by a single call. One unusual or disturbed call can therefore determine the entire value. Pairing the repeats does not make this single-call maximum suitable for reliable statistical testing.

Pooled p99.99 latency is not tested either. This value is calculated from all individual calls combined into one distribution. Because the individual calls, rather than the repeats, are the observations, it cannot be used with the paired repeat-level statistical test.

Finally, latency percentiles are not statistically tested after saturation. At saturation, the sender can no longer maintain the planned request rate and starts to fall behind. This is called *schedule drift*. The measured latency then includes both the actual service time and the time spent waiting because the sender has fallen behind. Therefore, these measurements cannot be used as a fair estimate of the service's processing time and are reported only as descriptive results.

3.7 Reproducibility and research artefacts

The code, raw logs, and rebuild scripts live in https://github.com/VienThanh12/Master-Thesis. Access to that repository is restricted during review, and the same package is submitted with this thesis. A reader with either copy can rebuild the results without access to Nokia's private code. The versions are driven entirely by the public fixed spec. The benchmarking harness, the fault catalogue and its runner, and the FFI prototype are part of that package. Each plot of measured data is rebuilt by one script from the raw logs. So is each quoted number in the results. The script is `bench/refresh_thesis.sh`. That claim does not cover the hand-drawn schematics, which explain a concept rather than report a result. And running the same scripts over another host's campaign rebuilds the artefacts by the same procedure, not with the same values.

The logs that script reads are part of the package. Each campaign has its own script and its own macro prefix, so a figure from one cannot reach a sentence about another. Each run's host setup is recorded and stored beside the campaign data. Nothing in the results chapters is typed from those files by hand.

4 Design and Implementation

Nothing is built until the spec is fixed. It fixes the public API, the UE state machine, the data layout and the steps on it, the locks and shard size, and the thread model. Both versions follow it. The test then times the language, not a gap of design or of skill. The spec is itself a result of the study. It is the main guard against the author favouring one language while writing.

One thread model is used on both sides: shared-nothing and thread-per-shard. Shared-nothing means one thread never reads another thread's records. Thread-per-shard means each slice of those records has one owner thread. That layout is known to change tail latency by itself [44]. The layout is held fixed as a fairness control. A production gNB-CU-CP schedules UE work differently. The id space is split into shards. Each shard is owned by one worker thread. That thread alone reads or changes it. So no lock guards context state. Callers send a request to the owning shard's bounded queue. They wait on a one-shot reply handle. The queue blocks when it is full. That is on purpose. A drop rule would hide overload. This study needs overload to show up as latency. The only shared mutable state either version allows is one atomic counter. No control flow depends on its value. Figure 4.1 shows the data flow. There is one multi-producer, single-consumer (MPSC) queue per shard. Its worker drains it first in, first out (FIFO).

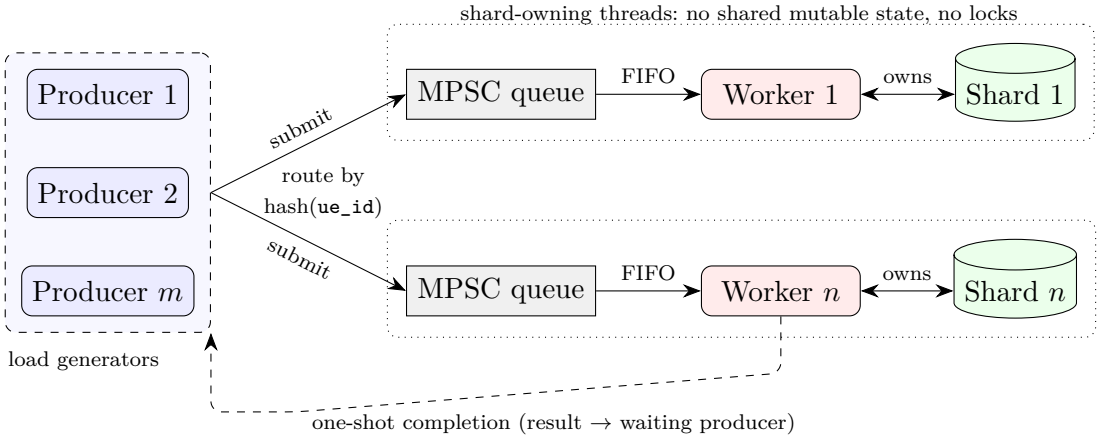


Figure 4.1: Shared-nothing, thread-per-shard architecture. Producers submit to a bounded MPSC queue per shard, and that shard's worker drains it.

Fairness is kept below the design as well as in it. Both builds link the same `jemalloc` and keep a system-allocator build as a control. Both compile at the same optimisation level, `-O2` for `g++` and `opt-level = 2` for Cargo, and both record latency through one

Rust

```rust
pub enum Request {
    // Create is left out here.
    Setup {
        ue_id: UeId,
        stamp_completion: bool,
        resp: CompletionTx<Completion<OpResult<()>>>,
    },
    Signal {
        ue_id: UeId,
        msg: SignalingMsg,
        stamp_completion: bool,
        resp: CompletionTx<Completion<OpResult<SignalingAck>>>,
    },
    // AddDrb, BeginReconf, CompleteReconf, AbortReconf,
    // Release, Lookup and Count are left out here.
    Stop,
}
```

C++

```cpp
using Response = std::variant<
    OpResult<void>,
    OpResult<SignalingAck>,
    OpResult<UeContextView>,
    OpResult<std::uint64_t>>;

struct Request {
  OpKind kind{};
  UeId ue_id = 0;
  CreateParams create_params{};
  SignalingMsg signaling{};
  DrbEntry drb{};
  ReconfigParams reconfig{};
  bool stamp_completion = false;
  std::promise<TimedResponse> completion;
};
```

Figure 4.2: The request type in each language, excerpted from `rust/src/message.rs` and `cpp/include/uecm/message.hpp` with comments and probe-only code removed, and with the `TimedResponse` struct left out. Each Rust variant except `Stop` holds only its own operation's fields and a reply channel typed to that operation's result. The C++ struct holds every operation's fields in every request, 160 bytes against Rust's 96. It replies through a `std::promise` whose `TimedResponse` wraps the `Response` variant of all results.

high-dynamic-range (HDR) histogram set up the same way [30]. Hashing is pinned on both sides. SplitMix64 assigns the shard [45], and FNV-1a, the Fowler–Noll–Vo hash [46], is used inside the map. C++'s default hash of an integer is often just the number itself, whereas Rust's is strong on purpose. Left as they came, the two standard libraries would have been the measurement.

Table 4.1: The forced-idiom log. Each row is one requirement of the specification and the way each language meets it.

Concern	C++	Rust
Result type	`std::expected<T, OpError>`	`Result<T, OpError>`
Request type	Tagged struct with every operation's fields, 160 bytes	`enum`, each variant with only its own fields, 96 bytes
Reply type	A `std::variant` of four results	Each operation's own result type
Bounded queue	Hand-written in 51 lines from a mutex and two condition variables	Standard `sync_channel(capacity)`
Reply handle	`std::promise` and `std::future`	One `mpsc::channel()` per request
Per-shard map	`std::unordered_map` with `FnvHash`	`HashMap` with `FnvBuildHasher`
Shared counter	`std::atomic<uint64_t>` by reference	`Arc<AtomicU64>`, cloned per worker
Not running	A call throws `std::logic_error`	A call panics
Teardown	The destructor calls `shutdown()`	`Drop` calls `shutdown()`
Optimisation	`-O2`	`opt-level = 2`

Where the languages cannot meet a need the same way, the gap is written into a forced-

C header and C++ host

```
typedef struct UecmService UecmService;
int32_t uecm_setup_connection(UecmService* h, uint64_t ue_id);

s = uecm_setup_connection(svc, ue);
check(s == UECM_OK, "setup_connection", s);
```

Rust layer

```
#[inline]
unsafe fn as_service<'a>(h: *mut UecmService) -> Option<&'a UecmService> {
    if h.is_null() {
        None
    } else {
        Some(&*h)
    }
}

#[no_mangle]
pub unsafe extern "C" fn uecm_setup_connection(h: *mut UecmService, ue_id: u64) -> i32 {
    match as_service(h) {
        None => UECM_ERR_NULL,
        Some(s) => status(s.inner.setup_connection(ue_id)),
    }
}
```

Figure 4.3: One of the 11 entry points of the C layer that lets C++ call the Rust service, excerpted without comments from `ffi/rust-uecm-ffi/include/uecm_ffi.h`, `ffi/cpp-host/main.cpp` and `ffi/rust-uecm-ffi/src/lib.rs`. The C++ host holds only an opaque handle. The Rust side checks that handle for null and dereferences it inside `unsafe` code, then calls the unchanged safe core, and `status` turns the core's typed error into an integer code. The layer marks 9 functions `unsafe`, and the core has 0 `unsafe` sites.

idiom log, which Table 4.1 lists. Two of those gaps were candidates for the overload lead of Section 5.3.2. The first is the bounded queue. Rust uses `sync_channel`. C++23 still has no matching primitive, so the C++ build implements the same blocking semantics with a mutex and two condition variables. The second is the one-shot reply handle. C++ pairs `std::promise` with `std::future`. Rust allocates a fresh `mpsc` channel for each request. Tests of those primitives and a replacement reply handle were run on an earlier host. They do not explain the remaining lead on this one once the extra delay samples are written after the interval. A third carries the core-operation result of Section 5.3.1: the per-shard container is each language's own standard hash map. A fourth is the request encoding. C++ has no payload-carrying enumeration in this idiom, so a request becomes a tagged struct holding each field any operation might need, 160 bytes against Rust's 96. The net result is a Rust core 20% longer in code lines. Its branching complexity is 13% lower [47]. The Rust library contains no `unsafe` at all, as Section 2.2 defines safe Rust.

Parity is tested and not assumed. Language-neutral vectors run in both versions. Their

logs are diffed operation by operation. A cross-diff cannot tell “both correct” from “both wrong in the same way”. A third version in Python acts as an independent oracle. An oracle is a third program, written on its own, whose answers the other two have to match. It has to reproduce all 223 hand-derived expectations before it may generate any of its own. A campaign of 9 000 generated operations found no mismatch. The C++ side runs under AddressSanitizer, ThreadSanitizer and UndefinedBehaviorSanitizer. The Rust side runs under Miri, an interpreter that checks `unsafe` Rust for undefined behaviour. A clean sanitiser result is bounded by the code that actually ran.

5 Experimental Setup and Evaluation

This chapter first describes the machines used in the experiment and the workloads that were run on them. It then presents the experimental results. Load comes first, as the service is swept from a light load to overload. The remaining research questions follow.

Several terms carry a specific meaning throughout this chapter.

Offered load is the rate at which the test sends requests to the service. For example, an offered load of 10,000 requests per second means that the test attempts to send 10,000 requests every second.

Saturation is reached when the service can no longer keep up with the offered load. Past this point, a higher requested rate does not raise the rate at which the service finishes work, because work begins to accumulate or the sender begins to fall behind.

Median latency is the latency at the middle of the measured observations. Half of the calls are faster than the median and half are slower.

Tail latency is the slow end of the latency distribution. It describes unusually slow calls rather than the typical call. Percentiles such as p99 and p99.99 name this part of the distribution.

5.1 Testbed setup

Some numbers in this chapter come from the tools and the code. Other numbers depend on the machine they ran on. The reported campaign comes from one bare-metal host, a single-socket AMD EPYC 7443P. Bare metal means the program runs on the machine itself, with no virtual machine in between. Each core runs one thread. Extra threads on a core are off. The host runs Ubuntu 24.04.4 LTS on kernel 6.17.0-23-generic. Linux can raise and lower each core's clock as load and heat change. The clock is held fixed. That clock setting is performance, and boost, a further rise of that clock, is off. The test may use only cores 2–17. There is no hypervisor. A hypervisor is the program that would run a virtual machine. The program runs on the host itself, not inside a virtual machine. On those cores the kernel turns off its timer tick (`nohz_full`). If left on, those ticks would

show up in the latency numbers. The study does not use `isolcpus`. That flag also stops the kernel sharing work across the allowed cores. Each test thread is placed on one core.

The compiler and build tools used in the experiment are not taken directly from the host system. Instead, both the C++ and Rust programs are built and executed inside the same container image. The image is based on Rocky Linux 8.10 (Green Obsidian) and provides `g++ 15.2.1` and `rustc 1.97.1`. This ensures that the build environment is controlled and reproducible rather than depending on the software installed on the host machine. The image specifies the major version of the GNU Compiler Collection (GCC) toolset, but not its exact point release. As a result, the compiler version used in this experiment is `g++ 15.2.1`. A container does not completely isolate the experiment from the host system. It still uses the host's Linux kernel.

Until late in the study, the load sweep was performed inside a Kernel-based Virtual Machine (KVM) running on an Intel(R) Xeon(R) CPU E5-2680 v4. The CPU core controls described above were not applied during this earlier sweep. The experimental notes record the hypervisor as kvm, the guest operating system as Rocky Linux 8.10 (Green Obsidian), and the guest kernel as 7.0.2-1.el8.elrepo.x86_64. Here, the guest refers to the virtual machine itself, while userland refers to the programs and libraries that run on top of the operating-system kernel. The earlier sweep showed substantial variation in the measurements. For one variant, the 95% bootstrap confidence interval for p99 latency had a width of 1480 μs. Across the four variants, the difference between the smallest and largest p99 values was only 293 μs. The uncertainty in one variant's p99 measurement was therefore larger than the spread among all four variants, so those ranked differences cannot be read with confidence. This earlier sweep is nevertheless retained because it used the same experimental design as the main experiment. That retention makes a side-by-side comparison possible. The purpose of this comparison is to examine whether the observed findings are consistent across different environments and therefore less likely to be specific to one particular machine or environment.

A third sweep was performed on a second bare-metal machine with an AMD EPYC 7443P processor. This sweep changed several experimental choices at the same time: the shard count, the set of offered loads, the number of repetitions, and the tools used. It therefore provides a broader robustness check. Specifically, it tests whether the findings remain similar when several experimental choices are changed together. It does not isolate the effect of any one of these changes separately.

Appendix A provides a field-by-field description of the three hosts and their corresponding experimental settings.

5.2 Workload and measurements

The work is made for the test. The steps it takes come from the 3GPP rules [1, 4]. A driver that always sends the same steps walks each UE through the connection lifecycle those rules define. The *signal* work first connects a set of UEs, then sends the same signal call again and again. That picks out the common path. The *mixed* work walks each UE through context creation, setup, signal, bearer setup, reconfiguration or handover, and release. A fixed random order makes both sides see the same counts of each call kind for a given seed. That is also a check that the two sides did the same work. Figure 5.1 shows the two loads and the steps the mixed one walks.

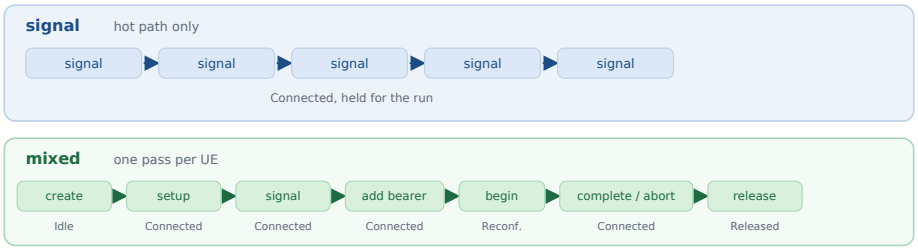


Figure 5.1: The two workloads, and the lifecycle the mixed one walks. Signal repeats the common path, and mixed walks each UE through the procedure sequence drawn below.

The test follows the 3GPP rules, which do not specify a particular mix of call types. In a real mobile cell, the handover rate depends on factors such as the location of the site and the time of day. There is no single publicly defined handover rate. Therefore, the call mix used in this study may differ from that of a real cell. However, the same mix is applied to all variants, so any difference in the mix affects all variants in the same way. It may change the absolute results, but it does not make the comparison between variants unfair.

The sender uses an open-loop approach, meaning that it continues to send requests according to the planned schedule even when the service becomes slow. If the producer falls behind this schedule, the test harness adds samples to the delay histogram for requests that should have been sent during the missed period. This follows the coordinated-omission correction approach used in HdrHistogram [29, 30]. In this study, this process is referred to as *histogram fill*, with up to 10 000 synthetic samples added for each completed call. These additional samples represent delays that could otherwise be missed when the sender falls behind. The term *histogram fill* is specific to this test harness and is not a standard benchmarking term.

Each load level is tested 20 times. The four language and allocator variants are interleaved within each run to reduce the effect of changes in the test environment. Each

reported value is the median of the measurements, together with a bootstrap 95% confidence band [42, 48].

For the resource-usage plots, the study uses a fixed load of 56 kilo-operations per second (kops/s), equivalent to 56,000 operations per second. This value was selected independently rather than taken from the load sweep. The initial rule was to use the highest load at which the p99 latency remained below 5 ms. However, on the virtual machine, this value was strongly affected by noise from the host system. On a quiet host, the load can be increased until the service is no longer able to keep up. Therefore, this measurement is used only as a check.

All four variants track the offered load through 392 kops/s. At 448 kops/s, Rust with `jemalloc` still tracks the offered load, but the other three variants do not.

5.3 Results

The safe-code results and most migration results do not depend on the host used. However, the speed and RSS results do. Unless otherwise specified in the figure captions, all measurements presented below were obtained on the host described in Section 5.1.

5.3.1 Below saturation (RQ1, RQ3)

This part compares the two languages while the service is still able to keep up with the offered workload. At the chosen load of 56 kops/s, all four builds remain sustainable. Each can process the incoming requests without falling behind.

At this load, the four median latencies differ by at most 1.6 μs, which is 2.2%. A paired test still finds a gap of 1.5 μs between the languages. Rust has slightly lower median latency than C++ under both memory allocators. Figure 5.2 shows the load sweep behind that number.

The same analysis extends to the higher tail percentiles. For one variant, the widest 95% band on its own p99 spans 0.8 μs. The four variants' p99 values differ by 1.8 μs. The uncertainty on one variant is therefore smaller than the gap between them. On the virtual machine the same two sizes were 1480 μs against 293 μs, the other way around.

Both p99 comparisons and both p99.9 comparisons remain significant after correction for multiple comparisons, 4 in all. That matches the median: the language gap is detectable at the typical call and in the upper tail.

Table 5.1 reports p99.9 and the maximum at the chosen load. The maximum is kept out of the tested family, as the pooled p99.99 is.

Table 5.1: How long one call takes at the chosen load, 56 kops/s, in microseconds. The median over 20 repeats sits above its bootstrap 95% interval. †The p99.99 column pools every repeat, so it has no interval. The maximum has an interval across repeats but stays outside the tested set, because one disturbed call can set it.

			(µs)		
Variant	p50	p99	p99.9	p99.99†	max
C++ (system alloc)	73.9	76.9	81.0	110.0	226.4
	[73.9, 74.0]	[76.4, 77.2]	[80.7, 82.0]		[221.1, 239.3]
C++ (jemalloc)	73.8	77.1	80.2	143.0	227.8
	[73.7, 73.9]	[77.0, 77.1]	[79.9, 80.9]		[217.2, 244.4]
Rust (system alloc)	72.7	75.3	77.1	87.9	100.3
	[72.6, 72.7]	[74.9, 75.6]	[76.9, 77.9]		[97.4, 160.8]
Rust (jemalloc)	72.3	75.7	77.8	81.3	104.4
	[72.2, 72.3]	[75.6, 75.8]	[77.6, 77.9]		[94.1, 141.8]

The far tail shows the same separation between the languages. The runs' histograms are pooled, which gives 3 360 000 to 3 360 063 calls per variant. The two Rust variants sit below both C++ variants. The widest pair differs by 62 μs. The slowest Rust variant still sits 22 μs below the fastest C++ one. A second bare-metal host at another shard count agrees, running 147–228 μs for the C++ variants against 88–115 μs for the Rust ones, and the ranges do not meet. A second limit does not come from the host. No host would remove it. An end-to-end round-trip spends only 0.250% of its time on the work the language does. The rest is passing the request to another thread, and that time is charged to the operating system. When the work inside the language is this small, time spent crossing from thread to thread dominates [49]. Figure 5.3 draws that split. This sweep cannot bound a language effect either way, which Figure 5.4 sets beside what the sweep can still show. Timed on its own, the core call is 1.6× faster in Rust. The test does not trace the cause. The stock hash map is one candidate. Compiler code generation may also own part of the gap.

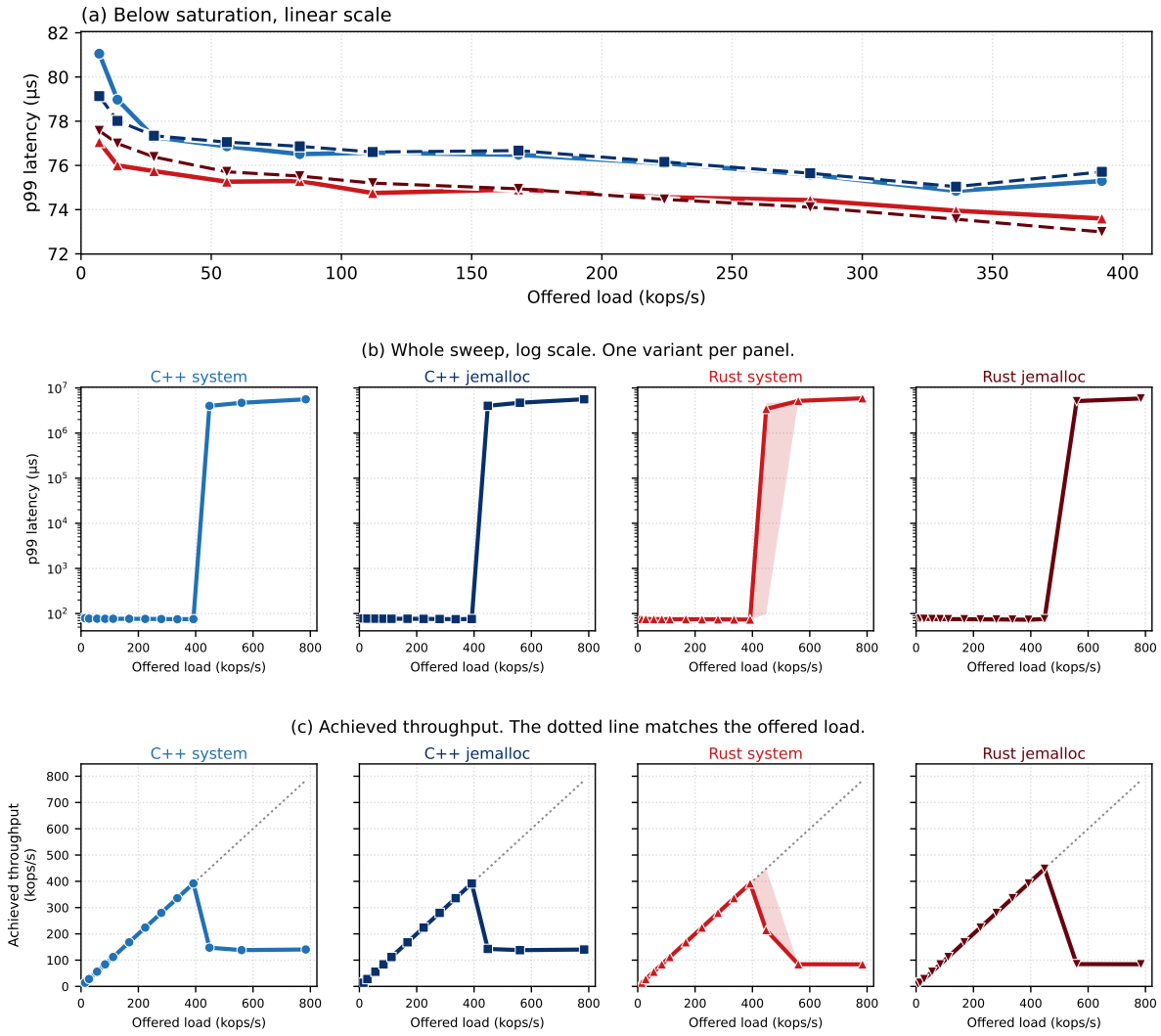


Figure 5.2: Panel (a) is p99 below saturation, on one linear scale, through 392 kops/s. The four variants sit within a few microseconds, so they are drawn together. In (b) and (c) each variant has its own panel. Panel (b) is the tail on a log scale across the whole sweep. Panel (c) is achieved throughput, and the dotted line matches the offered load. Lines are medians over 20 repeats, and the bands in (b) and (c) are bootstrap 95% confidence intervals.

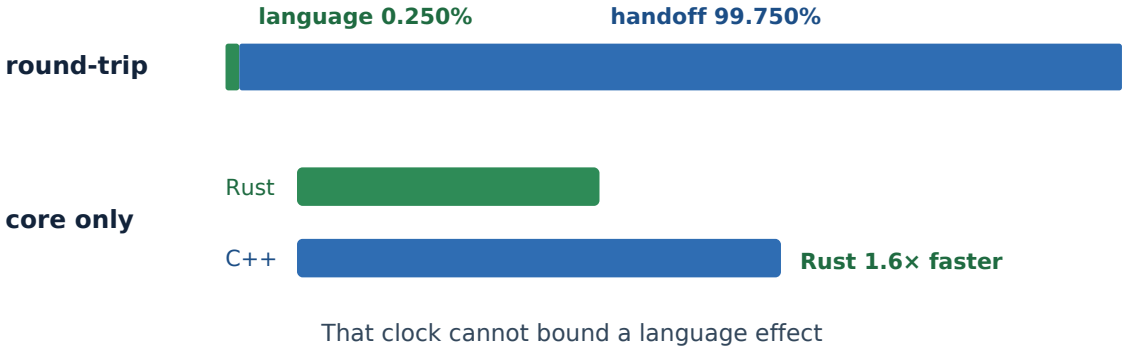


Figure 5.3: Where an end-to-end round-trip's clock goes. The language share is 0.250%. A clock that spends the rest on handoff cannot bound a language effect.

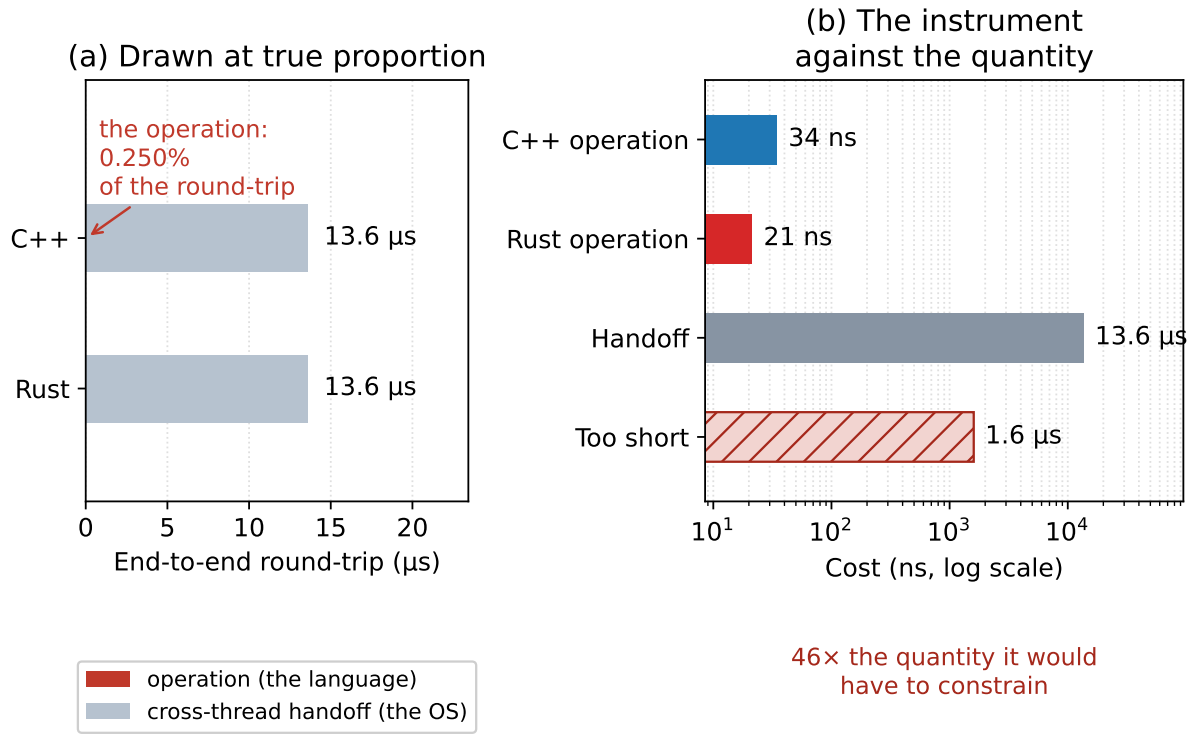


Figure 5.4: Where an end-to-end round-trip's cost goes. **(a)** The round-trip at true proportion. **(b)** Both operation measurements are too short for that clock to separate, by a factor of 46.

5.3.2 Past saturation (RQ1)

Past saturation Rust finishes more calls than C++ on both allocators. The lead comes from how each standard library hands a request to the worker and the reply back, not from the core operation. With several requests in flight, which language waits less depends on the allocator.

Each request has a scheduled send time. When the service is slow, some requests are still waiting to be sent even after their scheduled time has passed. Looking only at completed calls would miss these delays.

To capture them, the harness records additional delay samples for requests that should have been sent during that time. These samples are recorded after the timed test ends, so they do not add extra work during the test. The same set of test points is used for each of the 1120 runs.

What the sweep shows. Rust keeps up to 448 kops/s and C++ only to 392 (Figure 5.5). At 448 kops/s, C++ serves 416 kops/s and Rust serves 448. At the highest load, C++ finishes 0.95 of Rust's throughput on the system allocator ($p < 0.001$) and 0.92 under `jemalloc` ($p < 0.001$). These comparisons were not in the tests chosen before the experiment. Once throughput stops rising, CPU per finished call differs by a factor of 1.03, with C++ the higher one.

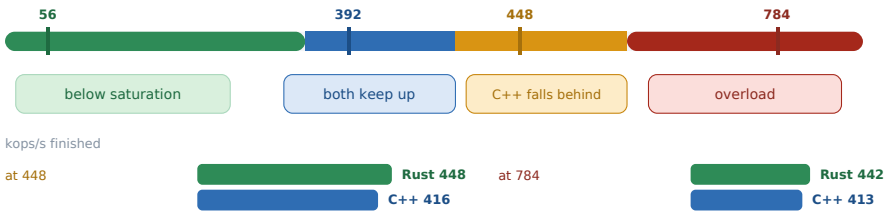


Figure 5.5: The figure shows four load points, ranging from light load to overload. These points are placed along the offered-load axis for easy reading. Their positions are not to scale. The tick marks show 56, the shared tracking load 392, the load where C++ first falls behind at 448, and the overload point at 784 kops/s. The bars show finished throughput, with extra delay samples written after the timed window.

Where the lead comes from. Rust finishes more calls past saturation, but not because its core call is faster. Each sender waits for its reply before sending again, so finished throughput tracks the full round-trip time, not the shard logic alone. That logic is at

most 0.250% of the trip. At the highest load, 57 to 63% of a call's CPU is kernel time spent putting threads to sleep and waking them. C++ spends more of that kernel time per call than Rust on 14 of the 14 load levels on both allocators, and 1.12 times as much at the top load.

Each follow-up run changes one part of the build and reruns the comparison. These runs were on further hosts, other machines with the same processor (Table 5.2). Changing the C++ map does not shrink the lead. Giving Rust the C++ lock and queue design removes it. Changing only the buffer behind that lock does not move C++. The lead therefore comes from how the queue is locked and signaled. On the C++ path the kernel sees 1.63 wake calls per request that find no sleeping thread, against 0.19 for Rust's channel. A final probe gives both languages one hand-written queue and reply around the same `ShardCore` (Figure 5.6). The lead then falls to 0.6% to 1.3% on two hosts. What remains is each language's own code around that shared path.

Table 5.2: Follow-up runs that locate the Rust lead past saturation. Each run changes one part of a build and keeps the rest. Hosts A to C are other machines with the same processor, listed in Appendix B. None of these runs is part of the Holm-tested family.

Host	What changed	What happened to the Rust lead
A	C++ uses an open-addressed map, as Rust does	Stayed. Rust ahead in 10 of 10 repeats
A	Rust uses the C++ queue design, one lock and two condition variables	Gone. Rust slower than with its own channel in 9 of 10 repeats
A	C++ keeps that lock with a fixed ring instead of a growable buffer	No change. C++ moved by 0.4%
B, C	Both languages use one hand-written queue and reply	Fell from 9.0% to 1.3% on host B, and from 7.8% to 0.6% on host C

Where a call's time goes. Timing each part of a call on host C shows where the round trip goes (Table 5.3 and the bars in Figure 5.6). The two handoffs take at least 97% of every call, split roughly in half between the queue and the reply. The worker's operation is at most 1.5% of a call. Through the standard libraries, 1.13 µs of Rust's 1.33 µs shorter call sits in the queue. With the same handoff, the queue and the reply take the same time in both languages within 0.10 µs. The stamps add two clock reads to every call, so whole calls are compared from the untimed runs.

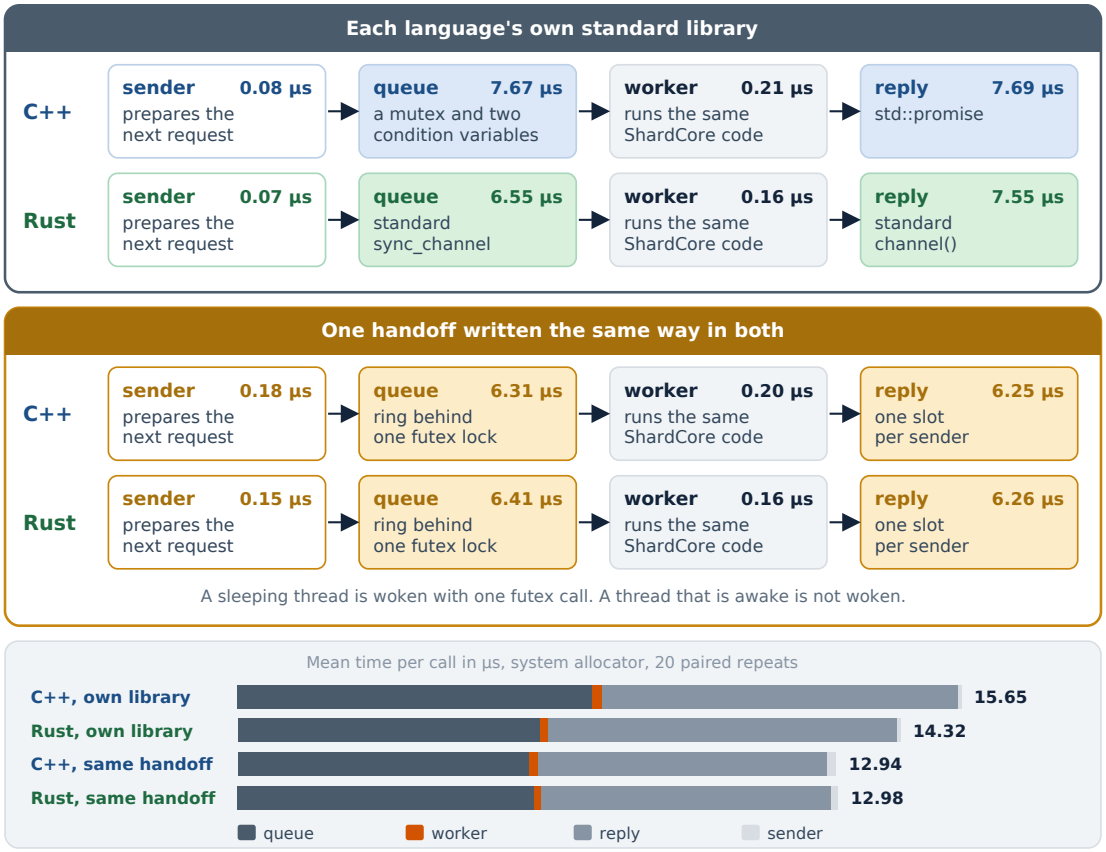


Figure 5.6: How a request reaches a worker, how the reply returns, and how long each part takes. The upper panel shows each language's own standard library, as in the reported service. The lower panel shows one hand-written queue and reply slot that both languages use. The worker runs the same `ShardCore` code in every row. Each box gives the mean time of its part, and the bars stack the parts into the whole call. The times come from host C, a further host with the same processor. This probe is outside the Holm-tested family.

Table 5.3: Mean time per part of one call past saturation, in microseconds, on host C (AMD EPYC 7443P), system allocator, median over 20 paired repeats. The queue runs from the sender handing the request over until the worker starts it. The reply runs from the worker finishing until the sender holds the result. The sender part is its own work before the next call. Rounded parts may miss the whole call in the last digit. This probe is outside the Holm-tested family.

	Own standard library		Same handoff	
Part	C++	Rust	C++	Rust
Queue	7.67	6.55	6.31	6.41
Worker	0.21	0.16	0.20	0.16
Reply	7.69	7.55	6.25	6.26
Sender	0.08	0.07	0.18	0.15
Whole call	15.65	14.32	12.94	12.98

Delay with several requests in flight. Past saturation the sender falls behind. The delay above then includes two things: the service time and the time the sender is behind schedule. That delay does not by itself show which service is faster.

A separate experiment on host D therefore times the work itself. Several requests can be in flight at the same time. Each request keeps its planned arrival time, and timing stops when the worker finishes it. The offered load is 3920 kops/s. It is the first measured load where every variant completes fewer than 99% of its planned requests within their arrival windows. That rule was fixed before looking at which variant did better. The experiment is not part of the Holm-tested family of the main sweep.

Table 5.4 gives the median for the signal workload and for the mixed lifecycle. With `jemalloc`, Rust has the lower median in every repetition for both workloads. With the system allocator, C++ has the lower median throughout. The difference comes from waiting to enter the queue, not from doing the work.

Table 5.4: Delay with several requests in flight on host D at 3920 kops/s. The median is the time spent waiting before entering the queue. In-service time is measured after queue entry until completion. This experiment is outside the Holm-tested family of the main sweep.

	Signal		Mixed	
	system	`jemalloc`	system	`jemalloc`
C++ median (ms)	200.7	121.9	199.8	120.4
Rust median (ms)	265.5	53.8	279.1	73.9
Rust lower	0/20	20/20	0/20	20/20
C++ in service (µs)	14.5	21.2	13.2	18.6
Rust in service (µs)	4.0	3.3	4.1	3.4

5.3.3 Resource use (RQ3)

The four variants have similar median latencies, but they do not use the same amount of RAM or CPU. The resource results show different patterns for memory and CPU, so the two are discussed separately.

Memory usage. Peak resident set size (RSS) shows two distinct effects: a *language effect* and an *allocator effect*. The language effect measures the additional peak RAM used by Rust compared with C++, while the allocator effect measures the additional peak RAM used by `jemalloc` compared with the system allocator. Table 5.5 reports both effects on all three hosts, and Figure 5.7 shows the results for the reported host at 56 kops/s.

The language effect is consistent across all three hosts: Rust uses more peak RAM than C++, and the difference remains within a few MiB. However, its exact size varies by host. In contrast, the allocator effect varies much more between hosts. On the virtual machine, for example, the allocator effect is about 13 times larger than the language effect. Thus, the relative size of the two effects is a property of the host environment, rather than a general property of Rust or `jemalloc`.

Table 5.5: Extra peak RAM on each host, in MiB. The language effect is the difference in peak RSS between Rust and C++, averaged across the two allocators. The allocator effect is the difference in peak RSS between `jemalloc` and the system allocator, averaged across the two languages. Each cell reports the range of these differences computed from the medians of the repeated runs. The reported host is measured at 56 kops/s. The other two hosts are measured at their respective steady-state loads.

	Reported host	Virtual machine	Second bare-metal host
Language effect	1.6–2.5	1.8–3.5	0.7–1.8
Allocator effect	1.5–2.3	32.5–34.3	1.9–3.2

The second bare-metal host provides more detail about the source of the additional Rust RAM. Table A.4 separates the process memory into allocator metadata, live application allocation, and allocator-resident memory. Compared with C++, Rust uses +2.07 MiB more arena metadata. Its live application allocation differs by −0.16 MiB, and its allocator-resident memory differs by −1.92 MiB. Although both are lower, the Rust process still has a peak RSS that is +1.74 MiB higher.

Therefore, the higher Rust RSS is not caused by Rust holding more live application data. Instead, it is mainly a fixed process-level cost.

CPU usage. CPU usage does not follow the memory pattern. At 56 kops/s, Rust uses at most 6.2 percentage points less CPU than C++. At 784 kops/s that ratio is 1.03, with C++ the higher one. A throughput lead remains, and Section 5.3.2 traces it to how the queue is locked.

Hardware counter reading. The hardware-counter measurements on the second bare-metal host cover only an isolated call. There is no queue, no offered load, and therefore no competition between requests for shared resources. The measurements show how much CPU work is needed to process one request when it runs by itself.

For this isolated call, Rust retires 11% fewer instructions and completes in 15% fewer CPU cycles than C++. It also has 29% more cache misses per thousand instructions. Thus, Rust completes the isolated call faster even though its cache-miss rate is higher.

These measurements show why Rust is faster than C++ when a single call is processed in isolation. However, the overload experiment is different: many requests arrive at the same time and must share the queue and its lock. Some requests therefore spend time waiting for the queue or for other threads to release the lock. This waiting does not occur in the isolated-core experiment and is not captured by the hardware-counter measurements. The overload experiment in Section 5.3.2 showed that the additional overhead comes from waiting for the queue lock and from the kernel time required for each handoff between threads. Therefore, the hardware-counter results explain the performance of an isolated call, but they do not explain the additional delays that appear when the system is overloaded.

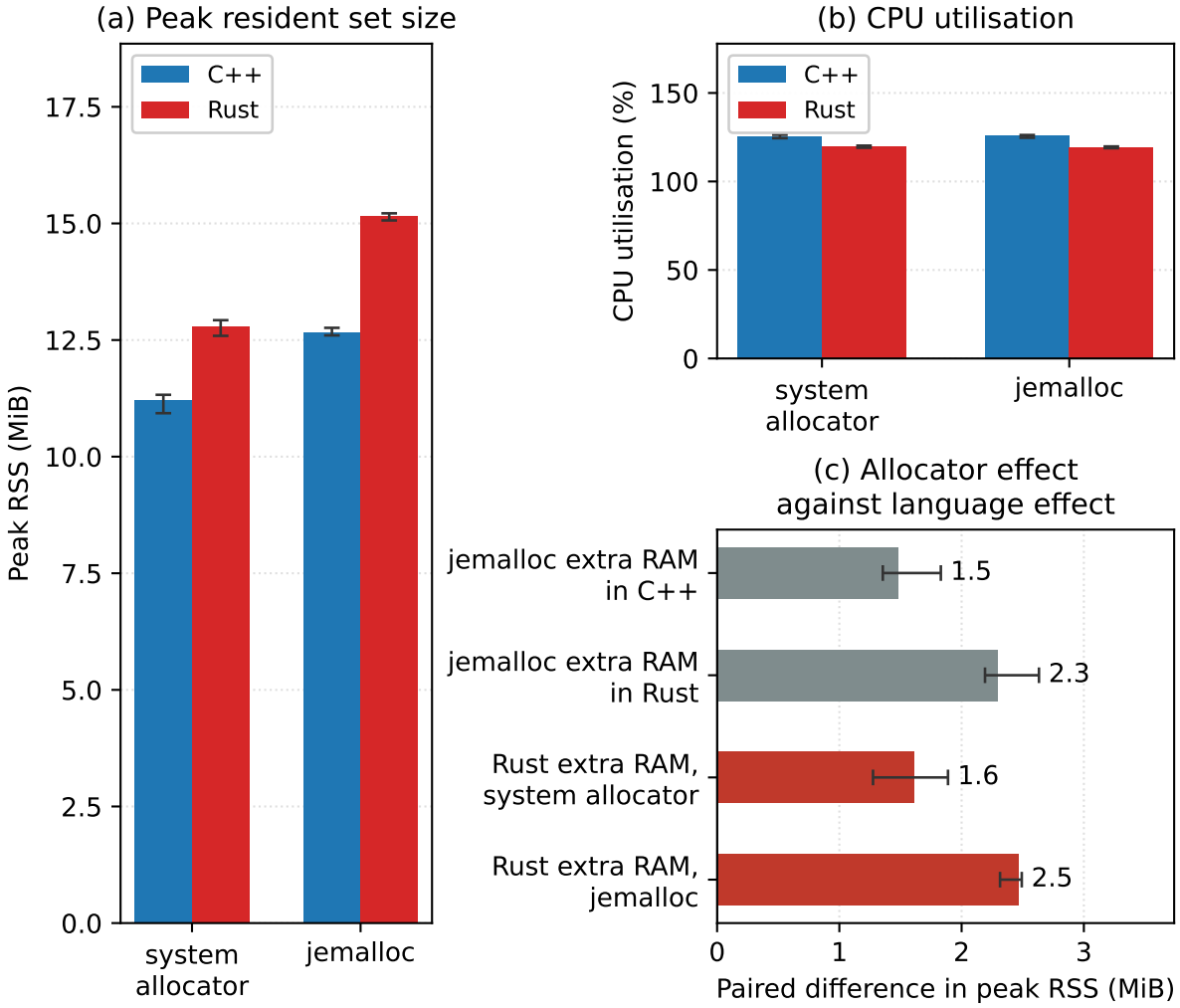


Figure 5.7: Resource use at 56 kops/s, grouped by allocator. Bars are medians over 20 repeats with bootstrap 95% intervals, and CPU in (b) is summed across threads.

5.3.4 Safety via fault injection (RQ2)

Safe code is evaluated by injecting 14 defect classes into the implementations. Seven of these defects fall within the safety guarantees that Rust is intended to provide, while the other seven represent defects that safe Rust is not expected to prevent. The first seven defects are prevented by the Rust compiler at compile time. In C++, the same defective code is allowed to compile and run. The defect will only become visible if the program actually runs the part of the code where the defect exists. Different defects require different detection methods, so no single sanitiser can detect all of them.

Use-after-move is an important example because it is prevented by Rust even though no sanitiser reports it. A moved-from C++ object remains valid, but its state is unspecified, so the program may continue running with the expected identity missing. There is therefore no invalid memory access for a sanitiser to detect, even though the application behaviour is incorrect (Section 3.2). Figure 5.8 shows the C++ program above the `rustc` error for its Rust version.

```
std::map<unsigned long, UeContext> store;
UeContext ctx{"244091234567890", 3};
store.emplace(1001, std::move(ctx));
// The acknowledgement is built from `ctx`, which no longer owns its identity.
std::printf("admitted UE '%s' with %d bearers\n", ctx.imsi.c_str(),
            ctx.drb_count);
```

```
error[E0382]: borrow of moved value: `ctx`
  --> faults/rust/use_after_move.rs:28:60
   |
20 |     let ctx = UeContext {
   |         --- move occurs because `ctx` has type `UeContext`, which does not implement the `Copy` trait
...
25 |     store.insert(1001, ctx); // ownership of `ctx` ends here
   |                        --- value moved here
...
28 |     println!("admitted UE '{}' with {} bearers", ctx.imsi, ctx.drb_count);
   |                                                  ^^^^^^^^^^^^^^ value borrowed here after move
```

Figure 5.8: Use-after-move, the one defect inside Rust's guarantee that no sanitiser reports. The C++ code at the top, from `faults/cpp/use_after_move.cpp`, moves the context into the store, reads the moved-from variable, compiles, and under each sanitiser prints `admitted UE '' with 3 bearers` with no report. Below it, `rustc` rejects the Rust version of the same lines with error E0382, shown in part with the path shortened. A static analyser such as `clang-tidy` can flag the C++ pattern, but the study ran none.

The seven boundary classes also compile in Rust, one of them only inside `unsafe`, so their faulty behaviour can still happen when the program runs. Some can nevertheless be detected by specialised tools. LeakSanitizer detects the reference cycle because the resulting memory leak is permitted by Rust's ownership model [50, 51]. UndefinedBehaviorSanitizer

detects the integer overflow in C++, while Rust defines the same operation to wrap when overflow checks are disabled. ThreadSanitizer detects the lock-order inversion in 3 of 12 runs, although this result varies between runs. The tool records lock-order relationships only when the relevant locks are successfully acquired. If a thread becomes stuck while waiting for the first lock, the other lock may never be acquired, and the tool then cannot see the full sequence it needs to report the deadlock.

The double-free inside `unsafe` Rust is not a separate defect class, but the same double-free error occurring in code where Rust's normal safety checks have been bypassed. The C++ implementation contains the corresponding error, so the difference between the two implementations is only in whether the compiler is able to prevent it. Miri detects the Rust double-free, but it cannot execute the C++ host code behind the foreign-function boundary.

Some concurrency defects cannot be found directly by the sanitiser tools. To investigate these cases, this study used Loom and Shuttle [52]. They systematically explore different possible orders in which threads can execute. These tools operate on small models rather than the full application. Each defect had to be converted into the tool's model and simplified to the smallest example that still reproduced the problem.

Rust's Resource Acquisition Is Initialization (RAII) prevents one common concurrency defect, forgetting to unlock a mutex. The lock is automatically released when its guard is dropped. RAII does not eliminate all lock-related defects. The two forgotten-guard cases in the study still caused the programs to hang, and neither tool reported these hangs. Figure 5.9 draws the whole list as an outcome matrix.

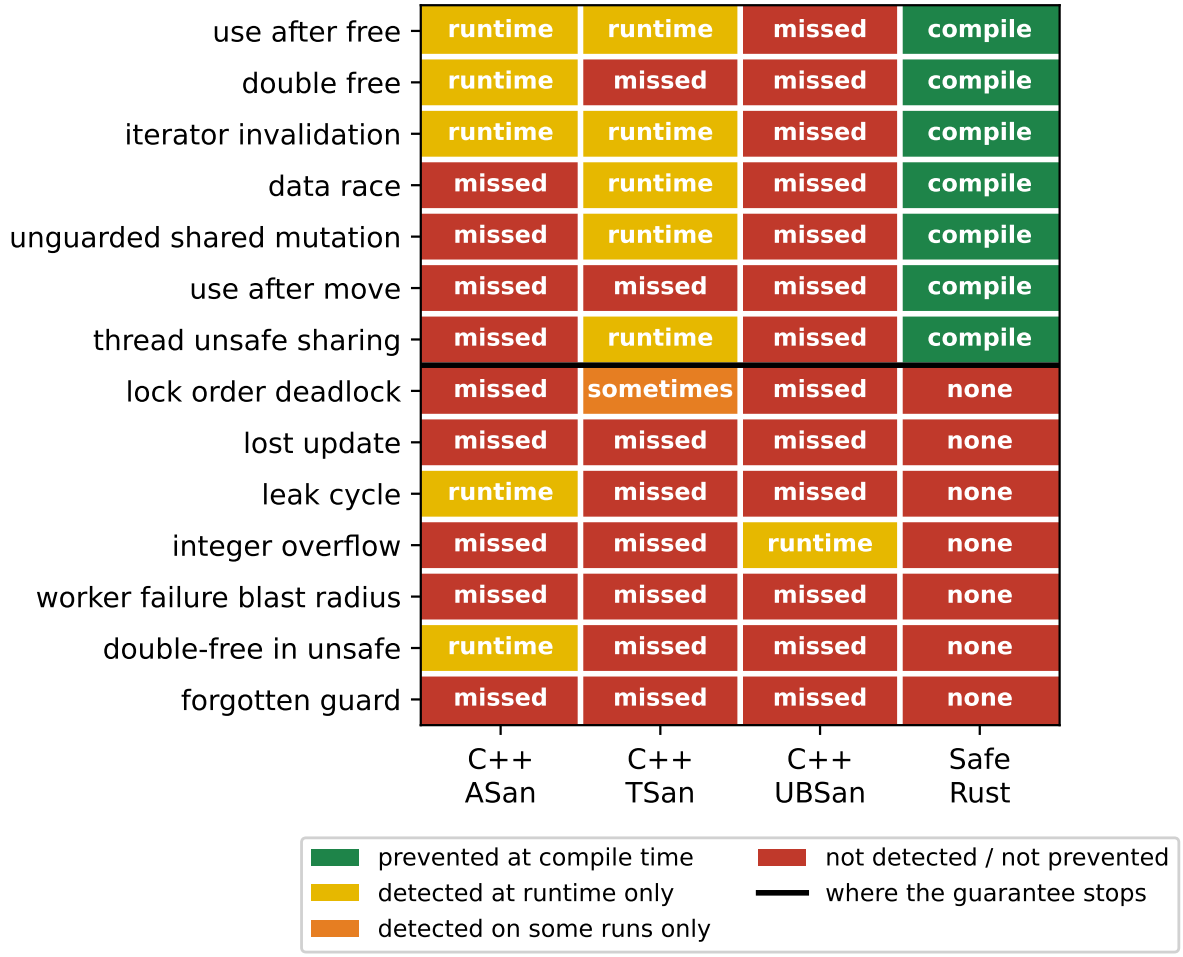


Figure 5.9: The fault-injection outcomes as a matrix, generated from the taxonomy log.

5.3.5 Migration (RQ4)

The C-ABI boundary between the C++ host and the Rust core is small. It consists of 11 C entry points implemented across 231 lines of Rust. All raw-pointer operations are isolated in 9 functions, leaving the main Rust core free of `unsafe` code.

No measurable runtime overhead is observed from crossing this boundary when comparing the same signal call made directly and the same call made through the C-ABI. Thus, any per-call overhead introduced by the boundary is bounded by 10.91% of the call cost in these measurements. Figure 5.10 shows the structure of this boundary.

Three practical costs of adopting Rust beyond the FFI boundary were also measured: build time, compiler cost for the project's own source, and shipped binary size. Third-party code accounts for 54% of the CPU time in a clean Rust build. For the project's own source alone, the Rust build requires 0.8 times as much CPU as the corresponding `g++` build.

The observed 6.8× binary-size difference is also strongly affected by the linkage policy. When the same linkage policy is used for both languages, the size difference changes and can even reverse. Finally, the Rust code used in this comparison contains no third-party crates and no `unsafe` sites (0 crates and 0 `unsafe` sites). These controls make the comparison more directly attributable to the language and build configuration rather than to dependencies or unsafe code.

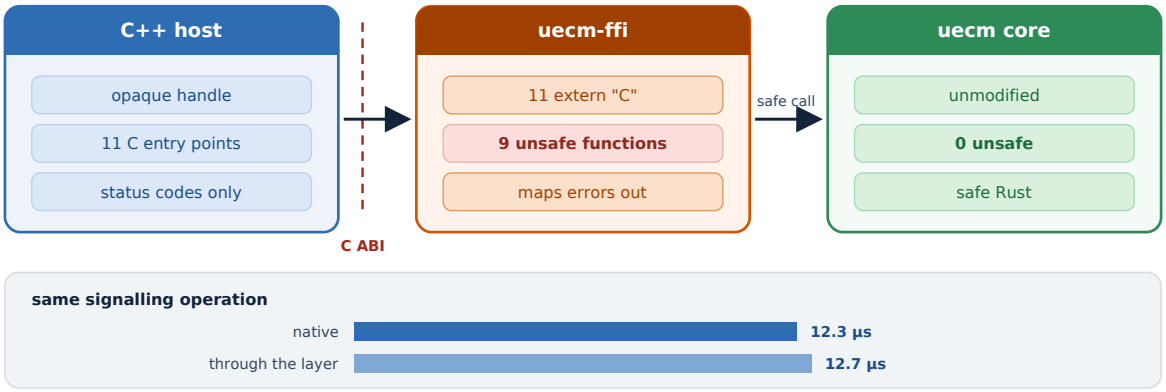


Figure 5.10: The layer where C++ calls the Rust service. The same signalling operation costs 12.3 µs directly and 12.7 µs through that layer, a paired difference bounded at 10.91% of that call. A do-nothing crossing costs 2.54 ns and bounds only itself.

6 Discussion

6.1 Summary of findings

- **RQ1 (latency).** At 56 kops/s, the variants show similar performance: their median latency differs by only 1.6 μs, while the pooled far tail favours Rust. Because only 0.250% of the end-to-end round trip is spent on language-specific work, the overall measurement cannot be used to establish a language-level performance effect. When the same core is timed separately, 1.6× higher performance is observed in Rust. At higher loads, however, Rust still completes more calls. This advantage appears to come from the inter-thread handoff rather than from the language itself. On further hosts with the same processor, giving Rust the C++ queue removes that advantage, and giving both languages the same handoff shrinks it to 0.6% to 1.3%. Past saturation the campaign delay is *schedule drift* as defined in Section 3.6, so it does not compare the services. Timing the work with several requests in flight provides a more meaningful comparison. As discussed in Section 5.3.2, the performance order in that case depends on the allocator.
- **RQ2 (defects).** The outcome is fully determined by the fourteen-class catalogue defined in advance and is consistent across hosts. All seven classes of errors that safe Rust is designed to reject result in compile-time errors. The remaining seven compile. They fail at runtime in both Rust and C++.
- **RQ3 (resources).** Rust has higher peak RSS under both allocators. The difference is 1.6–2.5 MiB on the campaign host, 1.8–3.5 MiB on the guest, and 0.7–1.8 MiB on the second bare-metal host. Below saturation, Rust uses slightly less CPU. After correcting for high-load effects, however, the throughput measurements do not support a general performance ranking between the languages.
- **RQ4 (adoption).** The foreign function interface layer contributes little overhead: its per-call cost is bounded by 10.91% of the corresponding call. Other factors, such as API design and learning effort, remain literature-backed adoption costs, but they are outside the scope of what this study measures directly.

6.2 Discussion on the research questions

RQ1 has two answers, depending on the workload level. At 56 kops/s, the campaign detects differences in the median, p99, and p99.9 latencies, while the pooled p99.99 latency favours Rust. However, the end-to-end measurements cannot be used to isolate a language effect because the core operation accounts for only 0.250% of the measured time.

After the system reaches saturation, the campaign delay mainly reflects scheduling drift rather than service performance. Additional delay samples are recorded after the timed window, so they do not add recording overhead while the timer is running. Therefore, these measurements should not be used to rank the services.

Completed throughput is a separate measure because it counts calls finished during the timed window. After saturation, Rust completes more calls with both allocators. This advantage appears to come from the handoff between threads, where C++ spends more kernel time per call. On a further host with the same processor, replacing the Rust queue with the C++ queue removed the advantage. With the same handoff in both languages, Rust stayed only 0.6% to 1.3% ahead on two further hosts. Most of the difference therefore comes from the queue and thread-handoff mechanism rather than from the Rust language itself.

On host D, a separate experiment times each request while several requests are in flight. It covers both the signal workload and the mixed lifecycle. For each workload, the measured delay includes the wait before queue entry and the time in service. Under this measurement, the performance order depends on the allocator.

The closest prior work reports a similar performance degradation above a threshold rate and attributes it to stackless `async/await` [34]. That explanation does not apply here because neither implementation uses an asynchronous runtime. Similarly, changing Rust's reply handle on an earlier host does not explain the remaining performance advantage observed here.

The throughput advantage should therefore not be interpreted as showing that one service has a higher maximum rate in all conditions. The original prediction attributed the difference to bounds and overflow checking [53, 54]. The results do not support that explanation. At nominal load, however, the median results are still consistent with previous evidence that Rust's safety checks are largely absorbed by modern out-of-order CPUs [12, 37].

For safe Rust code, some defects result in compilation errors and therefore never produce a program that can run. The seven classes of defects that Rust does not prevent are

therefore part of the result, rather than an additional warning. A compiler that rejects a program rules out a defect for every run of that program. A sanitiser can rule out a defect only for the executions covered by the sanitiser test. In other words, sanitisation improves the coverage of the tested executions, but it does not reduce the overall range of possible defects.

Figure 6.1 shows where the guarantees provided by each approach stop. Rust provides strong guarantees about memory safety and aliasing, but it does not guarantee correct lock ordering. For example, a `Mutex` guard whose lifetime is tracked by the borrow checker does not guarantee that two threads acquire multiple locks in the same order. Static deadlock detection in Rust remains an open research problem [55].

A lost update is similarly outside the protection provided by the borrow checker and is not necessarily detected by ThreadSanitizer. When every access is correctly synchronised, there is no unsynchronised access for the tool to report. A field study of 105 real concurrency bugs found that about one third of the non-deadlock cases violated the intended ordering of operations rather than the locking rules themselves [56]. Rust therefore removes some classes of concurrency defects but does not remove defects that arise from incorrect concurrent logic.

For this reason, adopting Rust should not lead to the assumption that "memory-safe" means "concurrency-safe." Such an assumption could result in insufficient review effort. In the corpus studied by Xu et al., all but one published Rust Common Vulnerabilities and Exposures (CVE) entry related to memory safety involved `unsafe` code. The exception was a compiler bug [57].

This is why keeping the core free of `unsafe` code is important. It also explains why the spread of `unsafe` code through dependent crates should be treated as a review concern [58].

One case shows that testing can also fail to cover a defect. A specification gap allowed a call to be sent outside the service's running window. The call passed every test, every parity vector, and all three sanitisers because the affected path was never executed. On the Rust side, the failure became detectable because of a simple bounds check, not because of a compile-time proof. This suggests that the useful engineering principle is "checked by default," rather than "ruled out at compile time."

The root cause of the problem also belongs to the testing method. A rule without a corresponding test or enforcement mechanism leaves each implementation vulnerable to its own failure modes.

RQ3 gives different results depending on the resource being measured. The RAM result is

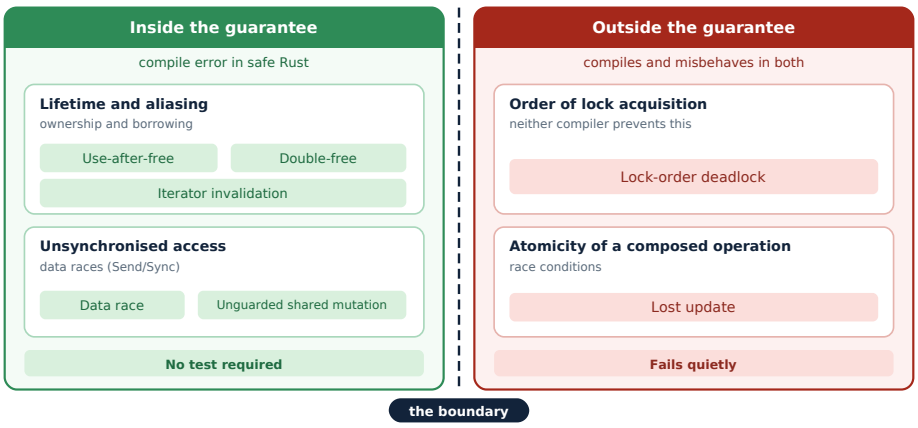


Figure 6.1: Where Rust's compile-time guarantee stops, with defect classes grouped by the mechanism that rejects them.

consistent across both allocators and on a second host using a newer compiler. Peak RSS measures the maximum memory used during the entire run, so the jitter that affects p99 latency does not directly affect this metric. On both bare-metal hosts the language effect and the allocator effect are of the same few-MiB scale. On the guest system, however, the allocator effect is an order of magnitude larger. The language effect shows on every host, at a size that varies by host. What depends most on the host is how it compares with the allocator's cost.

The second bare-metal host shows where Rust's extra memory sits under `jemalloc`. It lies outside what the allocator holds, mainly as a fixed cost of the process and not as more live data (Section 5.3.3). One related study reports the opposite result, with Rust using less memory and the difference increasing with concurrency [31]. These results can be reconciled if the implementations are viewed as using different concurrency runtimes rather than as direct comparisons of Rust and C++.

In that study, the C++ implementation uses Boost.Fibers, which allocates a stack for each fiber. The Rust implementation uses Tokio, where a task is represented by a small future stored on the heap. Therefore, the measured difference is partly a difference between two concurrency runtimes.

Two additional studies are consistent with the findings reported here [32, 59]. Below saturation, Rust uses slightly less CPU. After saturation, the CPU ratio reaches 1.03. This difference cannot be directly attributed to Rust's safe-code model. Nevertheless, Rust still maintains a throughput advantage.

RQ4 shows that the boundary between the two languages adds relatively little overhead. A C++ host can call an unchanged Rust core through a thin C-ABI layer, and the overhead

of a real signal call is bounded by 10.91% of the total call time. Previous work identifies design and learning as important human costs [60, 61], but this study does not measure their magnitude.

Keeping `unsafe` code inside the interface layer limits the amount of code that requires additional review, although it does not eliminate the possibility of memory corruption. The host and the Rust component share the same address space [62, 63]. The build-size and footprint results point in the same direction: organisations concerned about these factors should examine their dependency and crate budgets before attributing the cost to the compiler.

The main area where C++ has a clear advantage is the edit-and-compile cycle for changes that affect only a single source file. The 6.8× size result also shows that the relative size depends on the chosen linkage policy. The ordering changes when the same policy is applied to both implementations.

6.3 Revisiting the hypotheses

Section 1.4 defined three hypotheses before any experiments were run. Revisiting them after the experiments makes clear which parts are supported and which are not.

H1 H1 is not accepted as written. No equivalence margin was defined, so a statistically significant but small median difference cannot be used to establish performance equivalence. At 56 kops/s, the paired test detects a difference of up to 1.5 μs in the median and also detects differences at higher percentiles. The pooled far-tail result favours Rust, while the isolated core favours Rust.

Past saturation the campaign delay is *schedule drift* as defined in Section 3.6, so it does not rank the services. Timing the work with several requests in flight gives a more useful comparison, and the performance order depends on the allocator. Therefore, H1 combines round-trip latency, core performance, and overload behaviour into a single claim that the experimental method cannot support.

H2 H2 holds for the seven injected defect classes tested in this study. However, it does not hold under the broader interpretation suggested by its wording. The hypothesis is supported for memory safety and data-race freedom, but not for concurrency correctness in general. It also applies only to *safe* Rust code. The example involving `unsafe` code shows that the same class of defect can still occur when those guarantees are bypassed.

H3 H3 is supported: any per-call overhead is bounded by 10.91% of the signalling call.

This conclusion is based on the upper bound of the paired difference. A non-significant difference alone would not be enough to show that the overhead is small. The additional claim that the main costs are design and training is supported by the structure of the integration, but it was not measured statistically in this study.

6.4 Engineering guidelines for incremental adoption

The following guidelines translate the findings into practical recommendations. Each recommendation should be understood within the limits of the corresponding result.

- Select the first component to migrate based on the simplicity of its API rather than its size. A component with a complex interface can introduce greater integration and scheduling risks than a smaller component with a simpler interface. To improve safety, isolate the C ABI in a separate library and keep the core logic free of `unsafe` code. This approach helps prevent `unsafe` code from spreading throughout the dependency chain [58].
- Do not reduce the budget for concurrency review. Prefer a shared-nothing design where possible, regardless of the language. The strongest concurrency result in this study is primarily a property of the design, not of either compiler.
- Treat the learning curve as a real engineering cost and track it explicitly. Measure the component itself as well as the complete round trip. Test workloads beyond the point of saturation, where the campaign can no longer keep up with the offered load. Also audit every operation on the producer's timed path, including histogram correction.
- On the tested host, moving this work outside the timed window removed the large high-load latency gaps while leaving a smaller throughput advantage.

6.5 Where this study sits, and what it cannot support

The studies most closely related to this work are the five vendor-hosted theses discussed in Section 2.3. Their conclusions differ substantially from one another.

The study reporting the opposite peak-RAM result can be reconciled with the present findings if the implementations are understood as different concurrency runtimes rather than as direct language comparisons.

Four limitations remain particularly important.

1. The same author implemented both versions. This can introduce bias or differences in optimisation effort. A similar problem affected a previous comparison, where the result initially appeared comparable after code review but later showed Rust performing 85–91% worse after additional optimisation work [35]. A parity test suite establishes that the implementations behave similarly, but it does not establish that they have similar performance.
2. The end-to-end clock measures the system design as a whole, while the actual language-specific component represents only a small part of the measured time.
3. Extra delay samples are written after the timed window. In these runs that held only for C++. The Rust harness stopped its timer after the writing, so Rust's throughput past saturation is understated (Section 5.3.2). A throughput advantage remains even when the service can no longer keep up with the offered request rate. The difference in kernel time appears to explain part of this advantage on the tested host. A queue-swap probe on a further host supports the same reading, and so does a probe that gave both languages the same handoff.
4. Past saturation the campaign delay is *schedule drift* as defined in Section 3.6, so it does not rank the services. Timing the actual work with several requests in flight provides a more meaningful comparison, and the resulting order depends on the allocator. This experiment is outside the family of measurements used in the original campaign.

6.6 Ethical and professional reflection

This study uses no personal data, subscriber data, or operator secrets. All traffic is generated specifically for testing. The service is based on live code rather than copied production code, so the resulting artefacts can be shared publicly.

- **Claim discipline.** Avoid making claims that the data cannot support. A vendor could interpret the RQ2 result as permission to describe a migrated component as "safe." However, the seven defect classes that Rust does not prevent show why such a conclusion would be incorrect. A deadlocked control plane can cause an outage, while a silent lost-update bug can lead to incorrect mobility decisions. Rust prevents neither problem.
- **Evidence scope.** Tail-latency measurements from a single host provide a weaker basis for a production rollout than they may initially appear to provide. This is why the earlier guest-system measurements are retained rather than discarded.

7 Conclusion and Future Work

This thesis compared modern C++ and Rust on a UE Context Management service that keeps per-device state, as a gNB-CU-CP does. It asked how late the tail is, which defects a compiler can stop, what the safe code model costs, and what adding Rust one piece at a time costs. Both versions were built from one fixed spec and driven along the same code paths. That is what separates a language effect from a gap in design or skill.

Safety results. The safety mechanisms act at different stages. Safe Rust rejects all seven injected memory-safety and data-race classes before they become programs that can run. The same seven compile in C++, where a sanitiser can report a defect only if a test executes the relevant path. The other seven compile and fail in both languages. On two of them, LeakSanitizer and UndefinedBehaviorSanitizer flag the C++ version without a rewrite, which Rust's tools do not. Rust therefore rules out data races and aliasing errors, but not race conditions, lock-order errors, leaks, wrapping arithmetic, or the blast radius of a failed worker. It does not stop what an `unsafe` block lets through, such as the injected double-free. The two model checkers flag the concurrency cases only after each defect is recast into their parts.

Migration and build cost. On migration that layer is small. An unchanged Rust core can be driven from C++ across the C-ABI layer where the two languages meet. The C-ABI layer has 11 entry points in 231 lines, and its 9 raw-pointer functions contain all `unsafe` code. The core has 0 `unsafe` sites. The cost on a real signal call is bounded at 10.91% of that call. Prior work identifies API design and the learning curve as adoption costs, but this study does not measure their size. Most of the Rust build's extra cost is building crates, not the compiler's own work. The size gap is a linkage policy. The order rests on which matched policy is chosen. Where C++ leads is the edit and compile loop. It leads only for edits that stay inside one source file.

Performance and resources. On speed the data first splits by what is being timed. The end-to-end clock starts before a producer sends a request and stops after the worker replies. It includes the queue, wake-up, and core call. Only 0.250% of that round-trip is the core work the languages perform, so this clock cannot bound a compiler effect. A second measurement removes the queue and cross-thread handoff and times the core call alone.

The core is 1.6× faster in Rust in that measurement. At the chosen load of 56 kops/s the paired test finds 1.5 μs in Rust's favour. At p99.99 the slowest Rust variant sits 22 μs below the fastest C++ one. Past saturation Rust still finishes more calls at 784 kops/s. Extra delay samples are written after the timed window. CPU per completed call is 1.03. On a further host with the same processor, giving Rust the C++ queue removes that lead, so it comes from how calls pass between threads, not from the core work. Past saturation the campaign gives no language rank on that clock (Section 3.6). Timing the work with several requests in flight does rank the delay. The rank depends on the allocator, for signal and for the mixed lifecycle. The delay is the wait before queue entry. On RAM the Rust build carries a few MiB more fixed footprint under both allocators. It does so in the same direction and MiB scale on all three hosts timed. The allocator choice costs about as much on both bare-metal hosts, and an order of magnitude more on the guest this study used first.

Overall claim. Taken as a set, these support a narrow claim. It is for stateful, concurrent, partitionable control plane services of this kind. At the loads where the service still meets its sent rate, the choice is which classes of defect a team wants to make impossible to write. The other half is whether it will pay the design and training cost of getting them. Past that point the campaign gives no second language rank on the end-to-end clock (Section 3.6). The ranking from several requests in flight depends on the allocator, and that run sits outside the tested family. What the thesis adds is these three artefacts:

- a fixed specification
- a parity-tested implementation pair
- a timing harness that reaches past the saturation region

A split of the round-trip says how much of what it times the language can own. A defect list maps where the guarantee stops.

Future work.

- Vary how the kernel uses large pages for ordinary memory on one host with all else held fixed. All three hosts agree that the Rust build carries the larger footprint. They disagree about how that sits against the cost of the allocator. The guest put it an order of magnitude higher, and neither bare-metal host does. The large-page setting is the candidate this study can name but not test, because it was not the same on each host and only the system allocator variant gets it.

- Replay anonymised production traces to check the call mix, or extend the harness to further control-plane parts or the data-heavy layer-2 and physical layer (PHY) path. A part with truly shared mutable state would work the two sides in a way a shared-nothing design does not.
- Test Rust async runtimes against the thread-per-shard model to split the runtime from the language in the tail result [34].
- Time what a split of the two allocators in one process costs to price what keeping a core free of `unsafe` is worth [62, 63].
- Run a longer case study that tracks defect counts and review effort while adding Rust one piece at a time. That is what would settle the human cost this study can only name.

Statement on the Use of AI Tools

Tool. Anthropic's Claude Opus 5, Anthropic's Claude Opus 5.5, Cursor Grok 4.6, and Cursor Grok 4.7 are the generative AI systems used in preparing this thesis. All three ran through the Cursor development environment from June 2026 through the writing of this thesis. University of Helsinki policy forbids naming a language model as an author. They are not cited as a source, which is the author's own choice.

What the assistance covered. The models drafted and revised the running prose, and proposed the order of the arguments. They wrote and refactored both versions of the UE Context Management service. They also wrote the load generator, the fault-injection catalogue, and the FFI prototype. The analysis behind the bootstrap intervals and the paired permutation tests is their work too, as are the figure, build and LaTeX sources. The abstract is this programme's maturity test. University policy never permits a language model in one. In this draft it has not yet been rewritten unaided, and this statement will say so until it has been.

What the models did not settle. The research problem, the four research questions, the three hypotheses, and the study's main control are the author's. That control is one frozen language-neutral specification and one concurrency model, so language remains the only planned gap between the two versions. The author also chose to carry defect classes from outside Rust's compile-time guarantee, which is what keeps RQ2 falsifiable. The author directed every measurement and chose the machines each result names. Some runs on rented machines were started by the models over a remote login, under that direction. The quantities the thesis quotes reach the page as macros rebuilt from the raw logs. The readings are the author's too.

How the output was checked. All model output, prose and code, was read by the author before it entered the thesis. Every bibliography entry was resolved against Crossref, DataCite, or the issuing body. An audit then graded how far each claim could be traced. The first grade is the page or clause itself. The second is a publisher-deposited record that carries the claim. The third is confirmed metadata behind a paywall. An entry whose supporting passage was never opened is listed, not absorbed. The audit found four passages that overstated or misplaced their source. One was a quotation absent from the work it was attributed to. All four were corrected. Code was checked against a conformance suite whose transcripts are diffed line by line. It was also checked against a fault catalogue that fixes each expected outcome before the defect is injected.

Responsibility. The author has reviewed the whole of this thesis and is responsible for all of it. Where it is wrong, the error is the author's.

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Appendix A Host and campaign configuration

Table A.1 itemises the three machines the campaigns ran on. The first is the bare-metal host of Section 5.1 that every performance and resource table of the main sweep is drawn from. The second is the virtualised host it replaced. The third is a second bare-metal host with a different design. Section 5.1 compares that third host with the reported host. The resident-set decomposition belongs to the second bare-metal host and is reported in Section 5.3.3. The fault catalogue was compiled and run on the development machine, whose toolchain its own caption records. Of the adoption-cost axes only build time is host-sensitive, which is why that one is reported as a paired ratio while the footprint and dependency figures are deterministic. The three small paired experiments predate the move to bare metal and carry no environment record, so each supports a ratio between two variants measured alternately in one session. Section 5.1 points here for the field-by-field record of what each campaign captured. Every column reports what was measured, and the protocol they are read against is published as `bench/TESTBED.md`. Table A.2 gathers the configuration of the service, the workload and the campaign. Each adjacent column pair isolates one change, except where the Kernel and Toolchain rows record otherwise. Against the virtualised host, the design and container image match but the machine and `g++` point release differ. Against the second bare-metal host, the shard count, load grid, repeat count, and compiler differ too.

Table A.1: The three machines the campaigns ran on, each reported as measured. Every hardware, kernel and toolchain cell is generated from that campaign's own environment capture rather than restated from the protocol of `bench/TESTBED.md`. The operating-system row names the machine and, where a campaign ran in a container, the image's userland beside it, so the kernel row prints one value for both. The last two rows are prose, because what an instrument set is adequate for is a judgement and not a captured field. Rows where all three agree describe the source or the method rather than any machine. The *Short of protocol* marks are deliberate and are argued where they appear.

Aspect	Reported sweep	Virtualised host, same design	Second bare-metal host, different design
Host	AMD EPYC 7443P, 24 cores, 1 socket	Intel(R) Xeon(R) CPU E5-2680 v4 guest, 16 vCPUs	AMD EPYC 7443P, 24 cores, 1 socket
Operating system	Ubuntu 24.04.4 LTS. Image Rocky Linux 8.10 (Green Obsidian)	Rocky Linux 8.10 (Green Obsidian)	Ubuntu 24.04.4 LTS
Kernel	6.17.0-23-generic	7.0.2-1.el8.elrepo.x86_64	6.17.0-23-generic
Hypervisor	none. Clock setting performance, boost off, verified on the machine	kvm. Frequency and turbo host-controlled, unobservable from inside	none. Clock setting performance, verified on the machine
CPU steal	None: no hypervisor to lose a timeslice to	Not characterised. The capture's single one-second sample read 0, and no steal probe was run on this host	0 of 60 intervals on 16 pinned cores
Isolation	`nohz_full` 2-17. `isolcpus` none, omitted on purpose (§5.1)	`isolcpus` none, `nohz_full` none	`isolcpus` none, `nohz_full` none. *Short of protocol*
Large pages	madvise	always, and only the system-allocator variant inherits it	madvise
Toolchain	`g++ 15.2.1`, `rustc 1.97.1`, pinned in a container image	`g++ 15.1.1`, `rustc 1.97.1`	`g++ 13.3.0`, `rustc 1.98`, so a compiler change is tested too
Pinning	Cores 2–17, 7 shards, 7 producers	Cores 2–15, 7 shards, 7 producers	Cores 2–17, 8 shards, 8 producers
SMT	off, verified	Absent from the topology exposed to the guest, and decided on the host	off, verified
Repeats	20 over 14 loads, 1120 runs	20 over 14 loads, 1120 runs	15 over 8 loads, 480 runs. *Short of protocol*
Instruments	Counters, pooled HDR sidecars, jitter, allocator accounting	None available in the guest	Counters at two scopes, HDR sidecars, allocator accounting
Adequate for	The pre-committed campaign-clock results. Not overload service latency	Everything except the far tail below saturation	Confirming campaign-clock directions survive a change of design

Table A.2: Configuration of the service under test, the workload driving it, and the measurement campaign. Service parameters are fixed by the fixed specification, with one exception: the shard count is set to match the cores the host offers (Section 5.1). All of them, that one included, are identical in both languages. Campaign figures are generated from the measurement logs together with the tables that report them. The representative load is fixed rather than derived, for the reason Section 5.2 gives. The common tracking limit is the lowest of the four variant-specific limits. The variants diverge at the next load.

Parameter	Value	Note
Service under test, identical in both languages		
Shards, and worker threads	7	one thread owns one shard. Identical in both languages, which is what the fairness control requires (§3.5)
Queue bound per shard	1 024 requests	bounded and blocking, so overload is observable
Initial map capacity	4 096 entries	reserved so steady state does not rehash
Bearers per context	at most 8	
Signalling payload	64 bytes	
Context ceiling per shard	200 000	
Shard selection	SplitMix64, then modulo	pinned: diffuses sequential identifiers
Intra-shard hash	FNV-1a, 64-bit	pinned on both sides and checked bit-for-bit
Workload		
Workloads	signal, mixed	mixed walks the full RAN lifecycle
Active UEs	5 000	
Offered-load levels	14	up to 784 kops/s
Representative load	56 kops/s	fixed, see caption
Flat throughput	392 kops/s achieved	all four variants track through 392 kops/s. Rust with `jemalloc` also tracks at 448 kops/s. Past saturation, extra delay samples are written after the timed window (Section 5.3.2)
Load model	open loop, fixed intended schedule	latency measured from intended send time
Campaign		
Build variants	4	two languages × two allocators
Repeats per sweep point	20	variants interleaved within each repeat
Measured runs	1120	
Latency capture	HDR histogram, [1 ns, 3.6 s], 3 s.f.	same algorithm and configuration on both sides
Summary statistic	median with bootstrap 95% confidence interval	intervals reported in full, not as half-widths
Language comparison	exact paired permutation test	14 tests, Holm-corrected
Attainable p floors	0.0000019 / 0.000027	uncorrected / after correction

What each campaign recorded. Each of the 1120 runs of the factorial design in Table A.2 contributes one newline-delimited JSON record. Nothing is aggregated at capture time. A record carries the configuration (language, allocator, workload, shard and producer counts, target rate), the outcome (operations completed, errors, wall-clock duration, achieved throughput), and the latency distribution from the HDR histogram (mean, standard deviation, the quartiles, p50, p90, p99, p99.9, p99.99 and the maximum, in nanoseconds). It also carries a shorter block for inter-operation latency variation (mean,

standard deviation, p50, p99, p99.9, the maximum and a sample count), the resource counters from `getrusage` (peak resident set size, user and system CPU seconds, CPU utilisation), and the per-operation-kind counts. A `jemalloc` variant adds that allocator's own accounting. A system-allocator variant stores those fields as unavailable rather than as zero, on the same principle as the clock setting and the turbo state. The kind counts are what makes a divergence between the languages show up as a parity failure (Chapter 4) instead of a performance difference. Each run also persists its complete HDR bucket contents to a sidecar, so the repeats can be pooled before a far-tail percentile is read (Section 3.6). A separate file records `perf` counters at two scopes, the end-to-end path and the operation on its own.

The three small paired experiments have a different shape. For each of 15 interleaved repeats the decomposition records the batch-mean cost in both languages, once for the operation called directly and once through the threaded service. A third dataset of the same shape over 20 repeats records sustained throughput through the bounded queue and completion primitives alone. With 7 producers, Rust sustained 59 kops/s and C++ 55, which the test cannot separate from no difference ($p = 0.226$). The FFI comparison of Section 5.3.5 has that structure over 15 repeats. All three predate the environment record, which is why each is read as a ratio between its two variants.

Build time is the one RQ4 axis that is a timing measurement. It is recorded over 15 interleaved repeats at 14 parallel jobs on the development machine, alternating the toolchains so the ratio survives a change of machine even though the seconds do not. The static footprint and the resolved dependency graph are read from the linked binaries and the manifests.

The fault catalogue is categorical. Each of the 14 cases yields four tool logs (AddressSanitizer, ThreadSanitizer and UndefinedBehaviorSanitizer on the C++ program, and `rustc` on the safe-Rust counterpart) and one taxonomy row carrying the three sanitiser outcomes, the detection stage, the compiler result, and the expectation registered before the study. The two original cases that safe Rust does not stop yield a second table from the model checkers, and the `unsafe` one a third from the interpreter. The table and figure in Section 5.3.4 are generated from those logs.

Every load sweep writes the environment record of Section 3.7 beside its results, including the fields a given host does not expose. The clock setting and the turbo state are stored as unavailable rather than left out, so the limitation of Section 5.1 is a property of the dataset. The virtualised campaign shares the sweep's design and predates the later instruments. Its records carry the percentiles and the resource counters and nothing else. That absence

is the measurement-validity argument of Section 5.1, which is why the campaign is kept. The second bare-metal campaign carries the same record shape as the reported sweep, including the allocator accounting that lets Section 5.3.3 decompose the resident set. It runs a different shard count, load grid, repeat count and toolchain. Each campaign has its own generator and macro prefix.

The two timing checks each cover the same fourteen-load, two-language, two-allocator grid. Past saturation, extra delay samples are written after the interval. The processor is AMD EPYC 7443P, the pin is 2–17, and `nohz_full` is 2-17, matching the campaign's AMD EPYC 7443P and `2-17`. The earlier run that wrote none of the extra delay samples was not repeated here and remains an earlier host, so it cannot supply this campaign's percentiles. The overload percentiles of these checks are *schedule drift* as defined in Section 3.6. At the top load Rust is ahead on both allocators, at $p < 0.001$ on the system allocator and $p < 0.001$ under `jemalloc`. CPU per completed operation is 1.03. A throughput lead remains.

The tables that follow provide supporting measurements for the main text.

Table A.3: Inter-operation latency variation at a sustained offered load of 56 kops/s, in microseconds, as a median over 20 interleaved repeats with a bootstrap 95% interval. Measured within one producer and never across producers, since two producers share no schedule and a difference between them is not a variation either one experienced. Reported for completeness rather than as evidence: no research question names this metric, and it is absent from the paired family of Section 3.6.

Variant	p50	p99	p99.9	max
C++ (system alloc)	1.4	22.3	43.6	154.0
	[1.3, 1.5]	[18.6, 33.4]	[39.6, 47.9]	[148.9, 167.0]
C++ (jemalloc)	1.6	22.5	43.7	158.7
	[1.5, 1.6]	[14.3, 31.9]	[42.4, 46.6]	[148.2, 170.0]
Rust (system alloc)	1.5	26.9	40.7	56.9
	[1.3, 1.7]	[20.9, 33.7]	[38.7, 45.1]	[53.4, 107.5]
Rust (jemalloc)	1.6	20.9	44.9	63.4
	[1.6, 1.7]	[15.9, 26.1]	[42.1, 46.6]	[55.7, 108.9]

The allocation profile on the second bare-metal host was taken to name the call sites behind the resident-set difference of Section 5.3.3. Two departures from the campaign explain why it cannot. The first profile ran closed loop at 455 kops/s, past the flat stretch bracketed at 448–512 kops/s. Repeating it at a swept load below saturation shrank the discrepancy by most of its size and did not remove it. What remains is that profiling requires an allocator built to sample every allocation and record a backtrace, so the variant under profile is not the variant the campaign measured. Each symbolised profile

Table A.4: Where the peak resident set goes on the second bare-metal host, from `jemalloc`'s own accounting in each of the 240 runs of the two `jemalloc` variants. Medians in MiB over the steady-state loads, with the paired language difference in the last column. The two rows below the rule are the argument: the allocator holds *less* in the Rust build, and the process holds more, so the difference is outside the allocator. The final row is the quantity the other rows decompose. The derived row is the median of the per-run difference and not the difference of two medians, because the two figures it subtracts come from one process at one moment, so the column does not sum exactly.

Component	C++	Rust	Difference
Live allocation (`allocated`)	0.78	0.62	−0.16
In active pages (`active`)	1.43	0.79	−0.64
Arena bookkeeping (`metadata`)	4.19	6.26	+2.07
Resident in allocator mappings (`resident`)	11.02	9.10	−1.92
Outside those mappings (derived)	4.04	7.86	+3.82
Peak resident set (`getrusage`)	15.08	16.82	+1.74

puts almost all of its flat bytes on a frame that allocates nothing. In the C++ variant 98.4% sit on a deallocation symbol. In the Rust variant 100.0% sit on the allocator's own backtrace routine. Attributing the footprint would need a profile taken at a swept load, on a build whose peak RSS reproduces the campaign's ordering.

Appendix B Overload probes in detail

The probes of Section 5.3.2 ran on four rented bare-metal hosts, listed in Table B.1. Each is a single-socket AMD EPYC 7443P with SMT off, the performance governor and `nohz_full` on the measured cores, without `isolcpus`. Each ran the campaign's container image. None of them is the reported campaign host, and no probe is part of the Holm-tested family.

Table B.1: The further hosts behind the overload probes of Section 5.3.2.

Host	Where and when	What ran there
A	Singapore, 30 September 2026	The map swap, the queue swap, the ring buffer, and thread switches per call
B	Lithuania, 19 October 2026	The same-handoff probe, kernel calls per request, and the timer test
C	Chicago, 20 October 2026	Time per part of a call, and a repeat of the same-handoff probe
D	Chicago, 6 October 2026	Several requests in flight

The same-handoff probe runs the real `ShardCore` in one program per language, behind one queue and one reply slot per sender. Both versions make the same Linux `futex` system calls, and they wake a thread only after it has marked itself as sleeping. The same program can call the reported service instead, so within one binary only the handoff changes. Senders send back to back without a timetable, and no histogram is kept.

On the system allocator Rust stayed ahead with the same handoff in 17 of 20 paired repeats on host B and in 16 of 20 on host C. Under `jemalloc` on host B the gap fell from 11.8% to 1.4%, which the test cannot separate from no difference ($p = 0.174$). With the same handoff both languages make 3.19 system calls per request. Rust spends 0.12 µs less user time per call, out of a 12.6 µs round trip.

On host A all four builds of the queue swap switch threads 2.8 times per call at 56 kops/s. The C++ lock therefore adds kernel time to each handoff but no extra thread sleeps.

The timer test on host B set the cap on the deferred samples to zero. In that mode this removes only those samples. Rust's throughput at the highest load rose by 2.4% and was higher in 9 of 10 paired repeats, while C++ did not change. After the Rust harness was changed to stop its timer before writing them, the same test no longer moved Rust.

On host C the probe stamped each call as the sender handed it over, as the worker started and finished it, and as the reply arrived, all on one clock. The worker's two stamps are the reported services' own, so every build pays the same two extra clock reads. The parts add up to each run's time per call.